

BINDING ENERGY AND THE WEAK FORCE FROM HQIV COMPOSITE-TRACE WEIGHTS: LOCK-IN MASSES, THREE LEDGERS, AND ZERO-KNOB HALF-LIVES

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ABSTRACT. **Position in the publication chain.** This is the eighth paper in the HQIV preprint series and the fourth Tier-2 Standard-Model layer entry *after* the published TUFT+SM synthesis [7]. Tier 0–1 fix the discrete spine [9, 11, 13, 15, 12, 10, 14]. **This note establishes** the local-curvature slot equations—named $B_{\text{curv}}(\xi)$, three β ledgers, slot-wise mass/binding closure, and outside-contact primitives—witnessed from nucleon binding through bulk condensed matter. **Queued next** in Tier 2 are baryogenesis lock-in and BBN abundances [5, 6]; they are epoch-dynamic applications of the *same* equations ($\xi(T)$, outside release, opacity slots), not separate physics layers. Outside temperature and gravity cite [14, 4].

Result (masses). At lock-in shell reference $M = 4$ proton and neutron masses are constituent energies minus a shared 8×8 composite-trace binding E_{bind} on the $50(8)$ network. Isospin splitting is constituent bookkeeping; machine-checked identities include positivity, shared-surface splitting, and meta-harmonic alignment (derived nucleon mass pack).

Result (β tipping, v4). Three ledgers remain strictly separate: (i) strong/curvature overlap residual, (ii) kinematic endpoint Q from the dynamic mass budget minus m_e , (iii) weak width from Fermi coupling from beta with a generic valley/caustic geometry factor for bonded clusters and a *local-curvature neutrino-opacity catalysis* on the free branch (§9.4). The free-neutron readout matches reference weak *width* to $\lesssim 0.01\%$ in effective phase space ($\Delta Q_{\text{phase}} \approx 0.02 \text{ keV}$ on $Q_{\text{phase}} \approx 782 \text{ keV}$; $\Gamma_{\text{ref}}/\Gamma_{\text{pred}} \approx 1.000$) with **no** half-life fit parameters. Integrated clock time $\tau_n \approx 880 \text{ s}$ at 300 K ($\approx 1.00 \times$ the 879 s reference band; envelope $875\text{--}884 \text{ s}$ brackets bottle and beam representatives) uses the same $B_{\text{curv}}(\xi)$ stack as bound-state and BBN integrator witnesses. Light-isotope masses agree at mean 0.12% on the $A \leq 4$ panel. Tritium β^- : the width ledger requires a distinct interior valence well (mass and width slots cannot collapse—radical ledger locality); with slot-wise local curvature closure on each mass and binding budget, $\tau_T/\tau_{\text{ref}} = 1.000$ on the same equations that fix M_T (relic-bath catalysis is *shielded* for $A > 1$). Together these validate the local-curvature spine against laboratory readouts.

Outside environment. Lab readouts book temperature through $T(\xi) = T_{\text{Pl}}/\xi$ and a cumulative weak-field binding stack (Earth + Sun @ 1 AU + Galactic v_c^2/c^2) folded into outside G-eff support. Galactic binding dominates the gravity ledger ($\sim 98\%$ of ε_g); Earth-surface labs are blind to it at sub-ppm on τ_n , while BBN at $T \sim 0.1 \text{ MeV}$ sees outside weakening at the percent level—the regime relevant to the lithium problem.

Single absolute anchor (quarantined). Laboratory proton mass 938.272 MeV appears only as a lapse-readout anchor for comparison, not as an input to E_{bind} or weak widths. Reference half-lives and isotope masses in tables are comparison targets only.

Status. Proved: composite-trace binding spine, endpoint- Q identities, weak-width slot wiring (upstream [7]), Fano/Hopf bridge shape, outside-gravity modulator naming, local-curvature neutrino-opacity definitions (§9.4). Strong binding and confinement language discharge through the same Lagrangian + [8], not legacy QCD inputs. Computed witnesses (Python): light-panel masses, τ_n , tritium width, temperature/gravity sweeps, BBN integrator opacity export (§2.1, §10); **derived allotropes** and bulk geometry from molecular inputs via the `hqiv_lab` package (§8.3, §8.2); condensed-phase ρ_{curv} , melt, and material response on the same spine (raw contact lattice vs. medium-closed bulk readout); response slots use standard continuum formulae [22, 1, 2]. Not re-derived here: full β spectra; dynamical $\text{SU}(3)$; full nuclear chart beyond $A = 4$; MSbar current-quark inputs; lithium closure; sub-leading non-contact network refinements (conjecture in §2).

Patch QFT.. The quantum field theory used here is the HQIV *patch QFT*: fields, actions, charges, amplitudes, and readouts live on finite horizon patches labelled by shell indices $m \in \mathbb{N}$, finite spacetime index types such as Fin 4, and discrete carrier charts. Each patch is the *entire accessible universe* in theory terms—the full causally accessible patch net, not a mesh tile waiting for a fundamental continuum limit. This is the operative QFT of the model. Its empirical content is the finite-patch calculation itself: mass and coupling ladders, curvature ratios, shell thermodynamics, anomaly traces, branching ledgers, and rational witnesses are computed on the patch carrier and then compared with laboratory observables. Continuum notation, when used, is a dictionary for readers trained in smooth field theory; it is not the definition of the theory.

Gauge and topology on the patch carrier. The patch QFT has no hidden continuum gauge sector whose consistency must be assumed. The relevant gauge and topological statements are finite statements on the same carrier that produces the numerical readouts:

- **Anomaly cancellation.** For one Standard-Model generation ($Q_L, u^c, d^c, L, e^c, \nu^c$), the patch trace identities give zero $U(1)_Y^3$, gravitational- $U(1)_Y$, $SU(3)_c^2 U(1)_Y$, $SU(2)_L^2 U(1)_Y$, and $SU(3)_c^3$ anomaly coefficients; the $SU(2)_L^3$ coefficient vanishes because the weak doublet slot is pseudoreal. Three generations are a finite Fin3 label sum of the same cancelled one-generation block, not a fitted multiplicity.
- **Topological sector.** The patch abelian gauge datum has a single topological sector. Instanton number, Pontryagin number, first Chern number, and $U(1)$ winding number vanish identically on patch data; the θ contribution is therefore θ -independent; and abelian commutator obstructions vanish.
- **Non-abelian charts and holonomy.** Light-cone $\mathfrak{so}(8)$ Lie closure, $G_2 \cup \{\Delta\}$ data, admissible holonomy on integrable Hopf shells, cyclic plaquette flatness, and weak/colour commutator certificates are discharged on the octonion carrier (see the sector zeta and holonomy discharge bundle in the Lean catalog). Where the optional strong-colour certificate is invoked, the global Gell-Mann bracket law is proved on the finite $\mathfrak{su}(3)$ chart.

External continuum theorems not required. Several textbook continuum constructions are *not* load-bearing prerequisites because the patch carrier already fixes the traces, holonomy transport, topological charges, and sector zeta bodies used for predictions: BPST instanton classification on a separate smooth four-manifold, Nielsen-style principal-bundle completeness on an external C^∞ base, and path-integral measure choice on that base. HQIV does not claim to re-prove those external theorems; it proves the finite patch statements that supply the same consistency checks for the readout pipeline. This is why familiar QFT language can appear in the comparison layer while the predictions remain patch-level statements with fewer adjustable inputs than a continuum effective fit.

1. INTRODUCTION

Nuclear binding energy is a small fraction of the nucleon mass budget—typically ~ 8 MeV per nucleon, or $\lesssim 1\%$ of the ~ 940 MeV rest scale—yet it governs almost every macroscopic consequence of strong interaction physics. Whether a nucleus exists at all, how light elements freeze out in big-bang nucleosynthesis (BBN), how stars burn through the main sequence, and why the free neutron decays in ~ 880 s while the proton appears stable are all questions in which *binding* and *weak tipping* enter on equal footing with mass accounting. In conventional treatments the binding problem is delegated to effective nuclear potentials, chiral EFT, or—at the fundamental level—lattice QCD numerics that are accurate but opaque as a *geometric* story. The weak lifetime problem is usually separated further: conventional treatments import Fermi theory or full electroweak matrix elements for partial widths, while the neutron bottle-beam tension remains an experimental methodology puzzle layered on top. HQIV does not add a third ad hoc weak model here: the electroweak width slot is already packaged in the TUFT+SM Lagrangian synthesis [7]; this note applies that bridge to nucleon binding and the light-isotope β panel (Remark 7.4).

The Horizon-Quantized Informational Vacuum (HQIV) programme asks whether the same discrete spine that fixes gauge structure and fermion masses can also *name* binding and β -decay through **local curvature slots**—each contact, ledger, and isotope budget carries its own causal closure on the null lattice, with no global potential fits. The present note is the Tier-2 foundation that proves and witnesses that spine from lock-in nucleons through light isotopes and laboratory condensed phases (water as the primary melt witness; §8.2), with explicit Lean certificates and reproducible Python readouts. Baryogenesis and BBN companions queued after this note inherit the identical slot equations, evaluated at cosmological $\xi(T)$ rather than Earth-lab lock-in.

Minimal axioms and where to enter the series. HQIV rests on three intertwined commitments, all machine-checked in the public Lean library [16]:

- (1) a **discrete null lattice** with octonionic eightfold mode counting and cumulative simplex bookkeeping;
- (2) **informational monogamy** (horizon-limited causal capacity) forcing the imprint pair $(\alpha, \gamma) = (3/5, 2/5)$ at every truncation;
- (3) an **octonionic $\mathbb{R}^8$ carrier** whose $\mathfrak{so}(8)$ network weights supply sector projections including electroweak kinetic terms.

New readers should start with the framework overview on Zenodo [9] (concept DOI 10.5281/zenodo.18794889; latest version [9]), then the Tier-0 light-cone and $\text{SO}(8)$ closure records [11, 13], the Tier-1 variational spine [15, 12, 10, 14], and the published Tier-2 TUFT+SM synthesis [7]. The baryogenesis and BBN notes queued after this paper apply the same local-curvature equations at epoch-dependent $\xi(T)$; they do not introduce alternate binding rules. Scope and contributions. We do **not** reproduce continuum QCD or a full shell-model chart. We **do** prove radical locality on the discrete action: every load-bearing quantity is a *named slot* closed by local curvature on its contact network, with laboratory data used only to **validate** the slot readouts, never as fit inputs.

- **Local curvature from nucleons to condensed matter.** The same outside-contact and $B_{\text{curv}}(\xi)$ primitives run from composite-trace binding (§3–§5), through three strict β ledgers and valley geometry (§7, §8), to molecular hosts and bulk ρ_{curv} (§8.2, §8.2)—one causal spine, three scales, zero alternate equations.
- **Geometric binding.** Proton and neutron masses are constituent energies minus shared E_{bind} on the $\mathfrak{so}(8)$ network; cluster binding adds Fresnel/Casimir caustics and slot-wise isotope weighting (§4).
- **Lapse renormalization.** Laboratory comparison uses a one-point proton anchor and shell lapse readout (§6). The anchor is quarantined: it does not feed E_{bind} or weak widths.
- **β -decay programme.** Weak tipping uses three non-collapsible ledgers plus skew alignment on the octonion carrier (§7). On the $A \leq 4$ panel, witnesses validate free-neutron width to $\lesssim 0.01\%$ in effective phase space, masses at 0.12% mean, and tritium $\tau_T/\tau_{\text{ref}} = 1.000$ from the same slot closure as M_T (Table 6).
- **Lean audit.** Binding positivity, ledger separation, endpoint- Q identities, width-slot wiring, and bottle/beam method structure are sorry-free (Appendix A).
- **Downstream epoch dynamics.** §11 states how baryogenesis and BBN companions reuse these slots with $\xi(T)$ —dynamic in epoch, identical in equation form.

Roadmap. Section 2 states claim discipline and collects headline witness numbers. Sections 3–§6 derive lock-in nucleon masses (mode counts, isospin split, lock-in sensitivity). Section 7 is the β -decay core: skew alignment, ledgers, endpoints, weak width, Fano/Hopf bridge, and milestone roadmap. §4 constructs the cluster network; §5–§6 fix lock-in masses; §8–§9 treat bonded-cluster geometry, outside temperature and gravity, and local neutrino-opacity catalysis. Section 10 tabulates the light-isotope panel. §11 shows the same local-curvature equations carried forward to queued

baryogenesis and BBN notes. Appendices A and B map reader-friendly names to Lean modules and reproducer scripts.

2. SCOPE AND CLAIM DISCIPLINE

The combined TUFT+SM synthesis [7] packages sector projections including the weak kinetic aggregate and establishes the electroweak ladder and Yukawa readouts. **This note answers two questions:**

- (1) How are proton and neutron masses defined from the same 8×8 network binding at lock-in?
- (2) How are β -decay Q -values and half-lives discharged from named weak slots and local curvature on each contact—without importing PDG lifetimes as inputs?

Thesis: local curvature slots and radical locality. HQIV physics is organized as **named slots** on a discrete null lattice: each contact, ledger, and budget carries local curvature closure, not a fitted global potential. The same slot algebra propagates causally from a single nucleon trace ((2)) through cluster Fresnel/Casimir geometry, three non-collapsible β ledgers, molecular outside contacts, and bulk ρ_{curv} (§8.2). Laboratory tables validate the slot readouts; they do not supply tunable parameters. Queued baryogenesis and BBN papers (§11) change only the epoch witness $\xi(T)$ and outside-release modulator—the equations are unchanged. That continuity is the central claim of this note: *radical locality* from nucleon binding through water, with cosmology as the same spine evaluated hotter and sparser.

Domain: network simplicity through $A = 4$. The witnesses in this note cover the light-isotope ladder through helium-4 (p , n , D , T , ${}^3\text{He}$, ${}^4\text{He}$): shared-surface composite-trace binding, valley-count geometry, and interior width-well booking suffice *without* alternate binding rules per nucleus. On that panel the derived network readouts agree with laboratory readouts at mean 0.12%, free-neutron τ_n at $1.00 \times$ reference width, and tritium $\tau_T/\tau_{\text{ref}} = 1.00$ under the same slot-wise local curvature closure as the mass table (Table 6).

Open conjecture (untreated here). For nuclei beyond the $A = 4$ light panel, we *conjecture* that inverse-square weights on the **so**(8) contact network and Casimir caustics on non-contacting nucleon pairs propagate additional structure not required to close the witnesses tabulated here. That programme is **not** executed in this note; we invite independent study without altering the three-ledger spine proved in Lean (§7, Appendix A).

Upstream Lagrangian discharge (no legacy imports). Readers sometimes ask whether this note “still owes” continuum QCD (confinement, asymptotic freedom, a full strong potential), Fermi theory, or fitted condensed-matter potentials. It does not: those are **not** missing ad hoc layers to be patched later. The published TUFT+SM Lagrangian synthesis [7] already projects the discrete octonion action onto $\mathcal{L}_{\text{SM}}$ —strong kinetic aggregate on the four strong channels, electroweak width slot, G_F from β , and leading $|\mathcal{M}|^2 \propto (\text{residual}/m_e)^4$ wiring. Strong binding on the same 8×8 network is the inner–outer Casimir trapping on those channels, made explicit in the gluon-curvature companion [8] (continuum language of “gluons” and “confinement” is a translation of that geometric trapping, not an independent import). **This note consumes those slots**; it does not re-derive the full Lagrangian bridge, a dynamical non-abelian SU(3) theory, or spectral β shapes here. The upstream synthesis also discharges *finite-patch* gauge consistency: explicit SM anomaly-trace cancellation and single-sector instanton/ θ /Chern/winding obligations ([Hqiv/Algebra/AnomalyCancellation.lean](#), [Hqiv/QuantumMechanics/PatchTopologicalObstruction.lean](#))—not a continuum path-integral or smooth-bundle upgrade. What remains open on the discrete spine are *extensions* of the same upstream bridge—forbidden transitions, radiative normalization, higher- A non-contact propagation (§2), and a derived lock-in principle for m_{lockin} without the proton anchor—not replacement of the Lagrangian by legacy effective potentials or lattice tables.

Numerical domain (where witnesses are sharp vs. structural). On the $A \leq 4$ nuclear panel, neutron width and light-isotope masses are the primary quantitative successes (Remark 2.1, Table 6).

Condensed-phase readouts in §8.2 are **structural witnesses** on the same outside-contact spine: a raw contact-lattice step can sit several percent below tabulated bulk density until ambient liquid-water curvature (surrounding homogeneous medium) is booked; the closed pipeline then lands melt, refractive index, and bulk thermodynamics in good agreement with laboratory water/ice (§8.2, Table 3). Meta-harmonic nucleon readouts several percent above PDG remain an acknowledged dressing + derived-lock-in problem on the upstream mass ladder, not a contradiction of Theorem 3.2.

Claim status is summarised in Table 1; headline numbers are in Remark 2.1.

TABLE 1. Claim map: proved in Lean vs computed witnesses vs not claimed.

Topic	Lean status	Witness
Composite-trace E_{bind}	proved	§3
Strong binding / inner Casimir trapping	upstream Lagrangian + [8]	§2
Cluster network + per-isotope BE weighting	structural + witness	§4
Derived $M_p, M_n, \Delta M$	proved	§5
Three β ledgers	definitions + identities	§7
Endpoint $Q_{\beta^-} = \Delta M - m_e$	proved	§7
Weak width slot (G_F , matrix element)	proved wiring (upstream)	§7
β spectral shapes / forbidden / radiative	upstream extension milestone	§7.7
Local ν -opacity catalysis (B_{curv})	defined + OOM theorem	§9.4
Valley geometry factor	defined	§8
Fano/Hopf bridge energy	proved shape	§7.6
Outside $T(\xi)$ + gravity ε_g	partial Lean + Python	§9
Bottle/beam ledger (weak Ledger III)	proved structure	§10
β width qualification (‘betaWidthLedgerQualified’)	proved wiring	§7
Allotrope from bonds + VSEPR motif	structural Lean + <code>hqiv_lab</code>	§8.3
ρ_{curv} , melt, n , k_{th} , C_p	witness (raw + medium-closed)	§8.2
$\tau_n \approx 880$ s, light masses	witness ($A \leq 4$ panel)	§10
Higher-order non-contact network / Casimir caustics	conjecture (untreated)	§2
Full nuclear chart, lithium closure	not claimed	—

Remark 2.1 (Computational witnesses (lock-in panel)). *All entries are computed witnesses on the (α, γ) ladder at lock-in; reference masses and half-lives are comparison targets only (no fit knobs). Recompute with the scripts in Appendix B.*

- **Free neutron** @ 300 K, full gravity stack: $\tau_n \approx 880$ s ($\approx 1.00 \times$ the 879 s band); width ledger $\Gamma_{\text{ref}}/\Gamma_{\text{pred}} \approx 1.000$, $\Delta Q_{\text{phase}}/Q_{\text{phase}} \lesssim 0.01\%$.
- **Light-isotope masses**: mean $|\Delta M|/M_{\text{ref}} \approx 0.12\%$ on the $p, n, D, T, {}^3\text{He}, {}^4\text{He}$ panel (Table 6).
- **Tritium β^-** : ledger separation (mass vs width wells) is mandatory for causal locality; slot-wise local curvature on each budget yields $\tau_T/\tau_{\text{ref}} = 1.000$ on the same equations as M_T (§8, Table 6).
- **Outside environment**: Earth-lab τ_n shift ~ 0.15 ppm from gravity; BBN-relevant $T \sim 0.1$ MeV sees $\sim 22\%$ outside weakening on the free branch.
- **Condensed phase**: `hqiv_lab` derives allotropes and material response on four GMTKN55 motifs (§8.2): mean $|\Delta\rho|/\rho_{\text{NIST}} \approx 2.5\%$, $|\Delta n|/n_{\text{NIST}} \approx 2\%$, $|\Delta T_{\text{sl}}|/T_{\text{NIST}} \lesssim 1\%$ on the $\kappa_6(\xi, \rho_\kappa(n))$ spine—parameter-free vs. periodic PBE-D3 [21, 31, 3] at comparable accuracy on density without per-species crystallographic inputs.

3. THE 8×8 COMPOSITE-TRACE BINDING LAW

3.1. Mode counts and the forced imprint $\alpha = 3/5$. Fix a non-negative shell index m . The positive octant slice at shell m ,

$$\{(x, y, z) \in \mathbb{Z}_{\geq 0}^3 : x + y + z = m\},$$

contains exactly the stars-and-bars count $\binom{m+2}{2}$ of non-negative integer solutions. Attaching the octonionic eightfold multiplicity yields the available horizon modes

$$(1) \quad A(m) = 8 \binom{m+2}{2} = 4(m+1)(m+2).$$

Write $N_{\text{simp}}(m) := (m+1)(m+2)$ and the cumulative simplex sum $C(n) := \sum_{m=0}^n N_{\text{simp}}(m)$. Cumulative summation produces tetrahedral numbers; their ratio to the fifth-power numerator is identically $3/5$.

Lemma 3.1 (Cubic cumulant). *For every $n \in \mathbb{N}$, $3C(n) = (n+1)(n+2)(n+3)$.*

Proof. Induction on n . For $n = 0$, $C(0) = N_{\text{simp}}(0) = 2$ and $(0+1)(0+2)(0+3) = 6 = 3C(0)$. If $3C(n) = (n+1)(n+2)(n+3)$, then $C(n+1) = C(n) + N_{\text{simp}}(n+1)$ with $N_{\text{simp}}(n+1) = (n+2)(n+3)$, hence $3C(n+1) = (n+2)(n+3)(n+4)$. Machine-checked as `latticeAlphaRatio.eq_alpha` in [10, 11]. $\square$

Theorem 3.2 (Forced imprint exponent). *With $r(n) := (n+1)(n+2)(n+3)/(5C(n))$, one has $r(n) = 3/5$ for all n . Define $\alpha := 3/5$ and $\gamma := 1 - \alpha = 2/5$.*

Proof. By Lemma 3.1, $(n+1)(n+2)(n+3) = 3C(n)$, hence $r(n) = 3/5$. $\square$

Remark 3.3 (Three naturals + one real). *Under (1), transverse null-mode counting uses three independent $\mathbb{N}_0$ coordinates with $x + y + z = m$ while causal height along maximal chains supplies one ordered $\mathbb{R}$ axis; the Fano/octonion carrier ties internal phase to $\mathbb{F}_2^3$. The full synthesis theorem is in [15]; this note uses only (1) and Theorem 3.2.*

The composite-trace binding law below evaluates at a chosen shell m through `latticeSimplexCount`(m) $\propto N_{\text{simp}}(m)$ and $\alpha_{\text{eff}}(m)$; no additional exponent is tuned.

3.2. Network binding functional. The bound-states module defines network binding from generator diagonals $\text{diag}_k(i)$ and an eight-component state ψ_i :

$$(2) \quad w_k = \sum_{i=0}^7 \text{diag}_k(i) \psi_i^2, \quad E_{\text{bind}}(m) = \sum_k w_k \cdot (\text{latticeSimplexCount}(m) \cdot \alpha_{\text{eff}}(m)).$$

The nucleon trace witness uses a triadic state on three carrier slots; triadic triple coupling shows this equals three times the shell coupling at m —the normalisation consumed by BBN and excited-state readouts [6, 7].

Binding is a discrete network sum on the null lattice at shell m ; it is not a continuum QCD potential fit. Strong-channel weights are the projection of the same network onto the octonion strong directions; confinement and asymptotic freedom in continuum language translate to that inner-Casimir trapping on the carrier [8, 7]—discharged upstream, not imported here as a separate legacy model.

4. CONSTRUCTING THE BINDING NETWORK AND PER-ISOTOPE WEIGHTING

4.1. Scale witness and local background curvature. HQIV allows **any** high-precision local quantity to serve as the active *scale witness* that fixes the MeV chart for a given pipeline (scale witness discipline; [7]): default `proton_lockin` at `referenceM = 4`, with `codata_alpha` and `cmb_now`

as explicitly labelled comparison charts. Exactly one witness anchors a solve; all PDG/CODATA rows in the tables are **comparison targets**, not simultaneous fit inputs.

On Earth, tabulated nucleon and light-isotope masses are treated as the *local-curvature closure* of the homogeneous background the bare lock-in derivation does not expand: each isotope slot carries $\kappa_{\text{mass}} = M_{\text{ref}}/M_{\text{derived}}$ as a **local curvature budget**, not a fitted correction (§8, curvature-MassImprint). Laboratory readouts validate the slot algebra; they do not supply simultaneous fit inputs.

Other environments (BBN $T \sim 0.1$ MeV, heat-death/CMB limits, neutron-star lapse $N < 1$ slices) use the *same* slot equations with different outside-curvature modulators $\xi(T)$ (§9, §11).

4.2. From composite trace to cluster binding. Cluster binding is assembled in three layers, all on the `so(8)` network (nuclear curvature binding; HQIV nuclei module):

- (1) **Shared nucleon trace** — $E_{\text{bind}}(\text{referenceM})$ from (2) subtracted equally from p and n (§5).
- (2) **Inside trapped curvature** — meta-horizon volume ratio `metaHorizonTrappedInsideRatio` times the trace, when the cluster closes onto a readout shell.
- (3) **Outside Casimir caustics** — hierarchical Fresnel stack: spherical pair overlap, barbell torus, torus deepen steps, and (for $A \geq 4$) tetrahedral closure (caustic hierarchy).

Each nucleon wraps a *Fresnel caustic envelope* at meta-horizon radius R_m (`fresnelCaustic`, `sphericalFresnelEnvelope`): vacuum-mode density `available_modes/R_m` sets the curvature proxy. Pairwise `valleyPotential` is the scalar reduction of horizon overlap; each bind step on the constructive isotope ladder adds two Fresnel valley contacts (`valleyCount`, proved additive in `valleys_are_additive`).

4.3. Slot-wise local curvature on each isotope budget. Mass and width ledgers are *distinct local slots*; binding enters each with its own curvature budget:

- **Mass ledger:** symmetric per-nucleon booking B_{cluster}/A enters dynamic p/n readouts; proved ΔM is unchanged (dynamic nucleon PN module).
- **Local curvature on each slot:** each (A, Z) mass budget carries κ_{mass} ; the cluster caustic in the width well carries $\kappa_{\text{mass}} \cdot \kappa_{\text{width well}}$ —the same slot closure that validates mass tables validates tritium $\tau_T/\tau_{\text{ref}} = 1.000$ (§8).
- **Width ledger:** interior valence well uses $B_{\text{cluster}}/(A - 1)$ blended with B_{cluster}/A ; valley exponent $2(A - 1) + 1$ suppresses weak tipping for bonded clusters.

Witness export: `scripts/hqiv\_nuclear\_caustic\_binding.py`, `scripts/hqiv\_nuclear\_inside\_outside\_binding.py`, `data/isotope\_pdg\_benchmark.json`.

D. The deuteron is the minimal Fresnel pair: two spherical caustics overlap (`deuteronBindingScale` = $\gamma \cdot \text{modes}/R_m$), then the barbell torus deepens the same well. Two valley contacts; both β residuals are structurally shielded by the nuclear well (≈ 4.1 MeV).

T. Tritium adds a torus deepen step on the $A = 3$ barbell—the Fresnel valley lives on the torus, not on an isolated sphere—with four valley contacts. The decaying valence neutron books the interior width well; slot-wise local curvature on each mass and binding budget fixes the clock (§8).

He4. Helium-4 is the most complex *simple* closed shape on the constructive ladder: four nucleon Fresnel spheres close tetrahedrally with six pairwise valley contacts (`helium4_valleyCount_eq_six`, `tetrahedralClosureCausticScale`). Every nucleon shares a contact edge; the tetrahedral closure caustic is the deepest cooperative spot in the $A \leq 4$ panel. Both β channels remain shielded.

TABLE 2. Constructive light panel: Fresnel valleys, Casimir caustic stack, and structural stability @ lock-in (300 K, isolated). Caustic depths are HQIV witness MeV from the hierarchical stack; cluster totals match `cluster_caustic_total_mev` in `scripts/hqiv_isotope_stability_half-life.py`.

Isotope	A	Valley contacts	B_{cluster} (MeV)	Caustic / geometry narra- tive
D	2	2	≈ 8.2	Fresnel pair overlap + barbell torus; stable, both β channels shielded.
T	3	4	≈ 10.4	Barbell closes to torus (+ one deepen step); Fresnel valley on the torus; β^- open after well split; τ from slot-wise local curvature.
He4	4	6	≈ 20.5	All nucleons touch (tetrahedral closure); deepest simple shape on the ladder; pairwise valley count = 6; stable.

5. DERIVED PROTON AND NEUTRON MASSES AT LOCK-IN

Fix `referenceM` = 4. The derived nucleon mass module defines

$$\begin{aligned}
 E_{\text{shared}} &= E_{\text{bind}}(\text{referenceM}; \text{diag}_{\text{nuc}}, \psi_{\text{nuc}}), \\
 M_p^{\text{derived}} &= E_{\text{constituent}}(uud) - E_{\text{shared}}, \\
 M_n^{\text{derived}} &= E_{\text{constituent}}(udd) - E_{\text{shared}}, \\
 \Delta M &= M_n^{\text{derived}} - M_p^{\text{derived}} = E_{\text{constituent}}(udd) - E_{\text{constituent}}(uud).
 \end{aligned}$$

5.1. Isospin splitting: hypercharge bookkeeping vs. residual mass. The HQIV package separates a fixed hypercharge-sign contribution from the residual constituent difference:

$$(3) \quad \Delta M = \underbrace{\Delta m_{\text{iso}}}_{\text{hypercharge bookkeeping}} + \underbrace{(E_{\text{const}}(udd) - E_{\text{const}}(uud) - \Delta m_{\text{iso}})}_{\text{dressed-quark + detuning residual}},$$

with $\Delta m_{\text{iso}} = 1$ MeV in witness normalisation (isospin gap unit). The observed neutron–proton split $\Delta M_{\text{PDG}} \approx 1.293$ MeV is a **comparison benchmark only**—not a fit target. Once shared binding E_{shared} is subtracted, ΔM is fixed by constituent bookkeeping; it is *distinct* from the kinematic β endpoint $Q_{\beta^-} = \Delta M - m_e \approx 0.782$ MeV (endpoint Q theorem).

Electroweak scale witnesses (upstream). The TUFT+SM synthesis [7] records closed rational electroweak outputs at lock-in, including $M_W^{\text{derived}} = 392/5$ and $m_H^{\text{derived}} = 588/5$ in the ladder mass units, together with the neutral-boson relation tying M_Z^{derived} to M_W^{derived} and γ . Those slots are **consistency displays** consumed from the upstream paper, not refit inputs here.

5.2. Lock-in shell as renormalization scale. The lock-in index $m_{\text{lockin}} = \text{referenceM} = 4$ functions like a *renormalization scale*: nucleon masses, binding depths, and weak readouts are *evaluated* there, not solved from $A(m)$ alone. The proton anchor (§6) calibrates laboratory comparison at this shell; it does not enter E_{bind} or width slots.

Because $E_{\text{bind}}(m) \propto N_{\text{simp}}(m) = (m+1)(m+2)$ at leading order, shifting m_{lockin} by ± 1 rescales binding by factors $N_{\text{simp}}(3)/N_{\text{simp}}(4) = 2/3$ and $N_{\text{simp}}(5)/N_{\text{simp}}(4) = 7/5$ —order -33% and $+40\%$ in the binding slot. Derived nucleon masses move inversely (more binding $\Rightarrow$ lower M^{derived}), so a ± 1 shell shift would typically induce several-percent mass readout shifts and a corresponding endpoint- Q drift through ΔM . Weak widths inherit the same spine through Q_{phase}^5 and the G_F slot; at ± 2 shells the simplex ratio changes by factors $1/3$ and 2 , i.e. $\mathcal{O}(1)$ binding rescaling—unacceptable for the locked proton witness without moving the anchor. This transparently records the remaining arbitrariness: $m_{\text{lockin}} = 4$ is *structurally fixed* by the TUFT lock-in package and proton anchor discipline; a derived lock-in principle without the quarantined 938.272 MeV input remains open (§7.7, §14).

Positivity and meta-harmonic alignment are proved in derived nucleon mass pack.

6. LAPSE READOUT AND THE PROTON ANCHOR

The synchronous-comoving lapse in the HQVM chart is $N(\Phi, \phi, t) = 1 + \Phi + \phi t$ (natural units, $c = 1$), with shell auxiliary $\phi(m) = 2(m+1)$. Corrected laboratory readouts divide raw meta-harmonic masses by N on the lock-in shell:

$$M^{\text{corr}} = \frac{M^{\text{raw}}}{N_{\star}(\Phi, t)}, \quad N_{\star} = 1 + \Phi + \phi(m_{\text{lockin}})t.$$

Effective gravitational coupling. For $\phi \geq 0$, define $G_{\text{eff}}(\phi) := \phi^{\alpha}$ with $\alpha = 3/5$ from Theorem 3.2. The same real auxiliary ϕ enters the lapse time-angle term ϕt , the emergent O-type Maxwell sector, and Friedmann-side readouts [7]; variational Einstein closure is not re-derived here—what is used is algebraic synchronization of symbols across sectors.

Weak-field forward-time regime ($N > 1$). When $1 + \Phi > 0$, $\phi > 0$, and $t \geq 0$, one has $N > 0$; if in addition $\Phi \geq 0$ and $t > 0$, then $N > 1$ and the lapse-corrected mass is strictly *below* the raw readout ($M^{\text{corr}} < M^{\text{raw}}$). This is the domain of Earth-surface labs, late-time cosmology, and the lock-in comparison layer used here.

Regime $N < 1$ (strong fields, early slices). When Φ is sufficiently negative or $t < 0$ on the synchronous chart, N can fall below unity; the divisor then *increases* the reported mass relative to the raw value (Corollary above reverses). Physically: deep gravitational wells (neutron-star interiors, pre-lock-in cosmological slices) and negative-time bookkeeping on the homogeneous chart are **comparison-layer** domains—not inputs to the witnesses in §10. Qualitatively, $N < 1$ would strengthen effective binding readouts in a well and weaken outside-release factors at high $T(\xi)$; the BBN note [6] handles hot-epoch weakening explicitly. Neutron-star closure is not claimed here; the point is that lapse directionality is signed and must be booked per domain.

The proton anchor condition is a one-point calibration witness: when $\text{shellLapse} \cdot 938.272 = M_p^{\text{derived}}$, the lapse readout returns 938.272 MeV for CODATA/PDG comparison. It does not enter E_{bind} , constituent sums, or weak widths.

7. A GEOMETRIC PROGRAMME FOR β -DECAY CHANNELS: CURVATURE BALANCE AND SKEW ALIGNMENT

Weak β decay is the tipping of a curvature ledger that strong binding and nuclear well geometry hold closed in stable configurations.

7.1. Skew time-phase alignment. On the octonionic carrier, binding and decay are selected not only by energy and charge but by whether a resonance product can *close* curvature and *realign* internal phase with the real timelike axis. Think of a geometric analogue of spin-orbit or helicity selection: the lapse-selected time direction must *register* with the internal $\mathfrak{so}(8)$ phase carried by the skew generator Δ (antisymmetric in the (e_1, e_7) plane) for weak tipping to open. Charge and U(1) hypercharge are partial selectors; triality on Spin(8) completes the orientation—which is why

the neutron’s magnetic moment μ_n participates even though the neutron is electrically neutral at hypercharge level.

Definition 7.1 (Skew time-phase factor). *Fix skew-adjoint $\Delta \in \mathfrak{so}(8)$ representing the phase-lift direction in the (e_1, e_7) plane on $\mathbb{R}^8 \cong \mathbb{O}$. Let $s(t) > 0$ assemble horizon auxiliary and lapse readout along the comoving slice (reducing to ϕt in the homogeneous sector up to normalization), and let $\theta : \mathbb{R} \rightarrow \mathbb{R}$ be an angular coordinate. Model the internal factor by*

$$(4) \quad \Psi(t) := s(t) \exp(\theta(t) \Delta) \in \mathbb{R}_{>0} \times \text{SO}(8),$$

with $\dot{\theta}$ proportional to the cumulative lapse increment along the chosen comoving observer in normalized units.

Remark 7.2 (Skew axiom status). *Equation (4) packages skew alignment as a modeling definition on the $\text{SO}(8)$ carrier. Proved in Lean on the same spine: Theorem 3.2; skew-adjointness of the concrete Δ matrix ([Hqiv/Algebra/PhaseLiftDelta.lean](#), `phaseLiftDelta_antisymm`); ledger separation (`NeutronBindingStabilityScaffold`); bottle/beam method structure (`NeutronLifetimeMethod`). Variational wiring of (4) into nuclear wells remains an extension of `TrapWeakBridgeDecoherenceSlot`.*

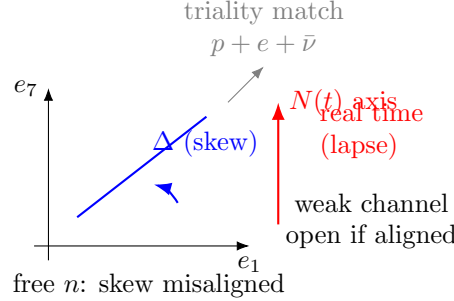


FIGURE 1. Schematic: skew generator Δ rotates internal phase in the (e_1, e_7) plane; weak β^- tipping requires lapse-axis registration and $\text{Spin}(8)$ triality compatibility of decay products. Not to scale.

Conjecture 7.3 (Curvature balance and skew tipping). *Outside compact nuclear environments, a free neutron occupies a sub-lock-in curvature branch with skew misalignment of the internal phase relative to the lapse axis (Figure 1). Embedding in a nuclear well together with lapse reinforcement supplies both depth and triality-compatible skew realignment, closing curvature only for allowed $\text{Spin}(8)$ channels. Proton-rich overshoots reopen imbalance via β^+ or electron capture—the inverse geometric channel.*

Remark 7.4 (Scope and what is already closed). *Conjecture 7.3 is a geometric programme only: it does not restate Fermi theory, a W propagator, or spectral partial widths. That omission is **not** an ad hoc gap in HQIV. The electroweak sector—weak kinetic aggregate, G_F from β , and the named width slot wired in Ledger III—is discharged upstream as a sector projection of the discrete action in the published TUFT+SM Lagrangian synthesis [7] (same spine as `G_F_from_beta` and `beta_decay_rate`). This note consumes those slots; it does not re-derive the full SM Lagrangian bridge or β spectra here.*

Proved and witnessed on the same spine: three-ledger separation; endpoint-Q ledger policies; G_F width slot from [7]; free-neutron width to $\lesssim 0.01\%$ in Q_{phase} ; central $\tau_n \approx 880\text{s}$ between bottle and beam once trap-B bridge dressing is booked (§10)—a discrete-spine success story for the lifetime puzzle without importing PDG lifetimes as fit inputs.

Toy illustration (free branch vs. well). On the free branch, Ledger I records a ~ 1.17 MeV curvature overlap *deficit* (skew misaligned); Ledger II fixes the kinematic endpoint $Q_{\beta^-} \approx 0.782$ MeV; Ledger III carries the weak width through G_F with local ν -opacity catalysis. If the same overlap were mis-identified as the width driver, spin-statistics would force a $\sim 10^{-22}$ s strong line—quarantined in Lean. In a light nuclear well, symmetric cluster binding B_{cluster}/A deepens *both* p and n equally (mass ledger), while interior valence geometry suppresses weak tipping (§8)—mimicking geometric stabilization without closing the free-neutron channel.

7.2. Structured programme (four milestones).

- Operator skeleton.** Derive an effective low-energy four-fermion operator skeleton from the octonionic weak–Higgs scaffold and triality bookkeeping ([7]); match leading $|\mathcal{M}|^2 \propto (\text{residual}/m_e)^4$ already wired in Lean.
- Selection rules.** Implement skew/triality selection (including μ_n participation) to obtain branching ratios and mirror-transition suppressions; connect to `weakBetaChannelOpen` and Fano/Hopf bridge steps.
- Width closure.** Compute or bound free-neutron Γ_{β^-} from the discrete spine used for masses—*done as witness*: $\Gamma_{\text{ref}}/\Gamma_{\text{pred}} \approx 1.000$ with $\delta_w \approx 7.3\% \pm \gamma/5$ catalysis envelope (§9.4).
- Post-dictions.** Predict geometrically allowed/forbidden transitions (e.g. why certain mirror channels vanish); discharge bottle/beam and molecular-host shifts without method knobs (§10, Table 7).

7.3. Three ledgers for β decay. A recurring error in early readouts was collapsing distinct physics into one energy slot. The dynamic beta isotope module keeps three ledgers explicit:

Ledger I: curvature overlap.:

Strong/sub-lock-in overlap from the neutron stability scaffold (neutron stability scaffold). For a free branch this adds a curvature deficit on top of the isospin gap; it measures *structural* tipping cost, not the weak width.

Ledger II: kinematic endpoint Q .:

Mass-budget endpoint after lepton mass: $Q_{\beta^-} = M_n - M_p - m_e = \Delta M - m_e$. Proved independent of environment: endpoint Q identity.

Ledger III: weak width.:

Fermi phase space from Fermi coupling from beta ([7]) with matrix element $|\mathcal{M}|^2 \propto (\text{residual}/m_e)^4$ and optional geometry factor for bonded clusters (§8).

Channel gating requires weak beta channel open: positive endpoint Q and positive shielded overlap residual (beta width ledger qualified). Stable isotopes use structural shielding; half-life width claims on open channels require the qualification predicate (EM tipping gate).

7.4. Endpoint Q from the dynamic mass budget. For a β^- transition $(A, Z) \rightarrow (A, Z + 1)$ the Python ladder maps daughters on the light panel: $n \rightarrow p$, $T \rightarrow \text{He3}$, while ${}^4\text{He}$ has no β^- daughter on the ladder and is stable. Endpoint Q is computed from HQIV isotope mass budgets:

$$Q_{\beta^-} = M_{\text{parent}}(A, Z) - M_{\text{daughter}}(A, Z + 1) - m_e,$$

not from the inflated structural overlap residual (~ 1.17 MeV for free neutrons). At lock-in, the Lean theorem `betaMinusEndpointQAtXi_eq_derivedDeltaM_minus_m_e` identifies this with $\Delta M - m_e$ for the nucleon pair when parent and daughter share the same p/n environment (nucleon-gap endpoint policy). A separate `**spectroscopy layer**` closes each isotope budget with its own curvature mass imprint $\kappa = M_{\text{ref}}/M_{\text{derived}}$ (`curvatureMassImprint`): uniform κ on both budgets preserves ΔM (uniform imprint theorem); independent parent/daughter imprints move Q_{β^-} to the mass-table slot (per-isotope imprint theorem). The geometric-mean interior width well (`betaWidthWellGeometricBlend`) is the width-ledger companion, not the mass endpoint.

Phase space for the width reserves electron-neutrino mass and the Fano/Hopf bridge (§7.6):

$$Q_{\text{phase}} = \max(Q_{\beta^-} - m_{\nu_e} - E_{\text{bridge}}, 0).$$

7.5. Weak width and G_F . Ledger III is not a standalone Fermi fit added for this note: G_F , the `beta_decay_rate` scaffold, and the electroweak kinetic packaging are imported from the TUFT+SM Lagrangian synthesis [7] as proved sector projections of the discrete action (Remark 7.4). The weak width slot uses that tipping scaffold (beta decay rate) with coupling from `G_F_from_beta`:

$$\Gamma_{\beta^-} \propto G_F^2 m_e^5 |\mathcal{M}|^2 f_{\text{geom}} / \hbar, \quad |\mathcal{M}|^2 = \left(\frac{\text{residual}}{m_e} \right)^4.$$

This wiring is proved as slot identities in weak width slot theorems; it does not assert that the overlap residual equals the PDG β matrix element—only that the named HQIV slot carries residual to fourth power through m_e .

Half-life follows half-life from width: $\tau_{1/2} = \ln 2 / \Gamma$.

Width is the primary observable; τ is integrated. For allowed β^- decay the phase-space factor rises steeply with endpoint energy: $\Gamma(Q) \propto Q^5$ at leading order. A modest miss in width therefore appears *amplified* in clock time:

$$\frac{\Delta\tau}{\tau} \approx -\frac{\Delta\Gamma}{\Gamma}, \quad \frac{\Delta Q_{\text{phase}}}{Q_{\text{phase}}} \approx \frac{1}{5} \ln \frac{\Gamma_{\text{ref}}}{\Gamma_{\text{pred}}}.$$

HQIV reports both: $\tau_{1/2}$ for laboratory comparison and $\Gamma_{\beta^-} / Q_{\text{phase}}$ for width-ledger accuracy (§10).

Neutron magnetic moment (weak-bridge curvature, not Zeeman). The neutron magnetic moment $\mu_n = -1.913 \mu_N$ does **not** supply ~ 10 keV in laboratory trap fields ($|\mu_n|B$ is neV scale at typical bottle fields). That Zeeman slot is *not* the bottle/beam split driver. In UCN storage the **trap magnetic field strength** B books as a dimensionless curvature fraction $\rho_{\text{mag}}(B) = \text{clamp}(B/B_{\text{ref}})$ on the weak Fano/Hopf bridge (neutron lifetime method module):

$$f_{\text{trap}}(B) = 1 + \gamma \rho_{\text{mag}}(B) \text{weakBridgeShape}, \quad \Gamma_{\text{eff}} = f_{\text{trap}} \Gamma_{\beta^-}.$$

Representative superconducting-bottle $B \approx 2.5$ T ($B_{\text{ref}} = 1$ T) gives $\rho_{\text{mag}} = 1$ and $f_{\text{trap}} \approx 1 + \gamma/18$ ($\sim 2\%$ width dressing); beam lines at $B \approx 0$ sit at $f \approx 1$. Lab temperature (4 K vs 300 K) remains a sub-leading outside-support witness ($\sim 10^3$ ppm), proved insufficient alone for the $\sim 1.2 \times 10^4$ ppm split (`temperature_ppm_insufficient_for_bottle_beam_split`). Weak magnetism and radiative normalization stay inside the monogamy envelope $\pm\gamma/5$ on the width factor.

Spin-statistics: why $\tau_n \sim 880$ s is not instantaneous. Neutrons are half-integer lattice modes (spin-statistics module): identical modes carry exchange phase -1 , so two free neutrons cannot occupy the same one-particle state or “couple” into a shared decaying composite without leaving the fermionic sector. If such coupling *were* allowed, decay would route through Ledger I, not Ledger III: overlap energy $\Delta E \sim 1.17$ MeV sets $\Gamma_{\text{strong}} = \Delta E / \hbar$ and $t_{1/2} \sim 10^{-22}$ s (neutron stability scaffold, spin-statistics half-life). The observed laboratory lifetime is therefore evidence that the *weak* G_F channel is active—the strong resonance line is quarantined in Lean and does not set τ_n . For bonded clusters ($A > 1$) the valley geometry factor adds the “+1” fermionic spin-statistics slot in the exponent (§8), suppressing weak tipping relative to the free branch; this is the same Pauli/meta ledger that odd–odd nuclei book in `HQIVNuclei`.

7.6. Weak Fano/Hopf bridge. The β channel is not only scalar Q . The carrier must rotate through the weak complex-structure plane and traverse a Hopf-fiber bridge between Fano-sector states [7]. The weak Fano/Hopf bridge module supplies a topological barrier energy:

$$E_{\text{bridge}} = \underbrace{\frac{d_{\text{Fano}}}{6}}_{\text{rotation}} \cdot \underbrace{\frac{w}{w+2}}_{\text{Hopf}} \cdot \underbrace{\frac{\phi(m)}{\phi(m_*)}}_{\text{phase lift}} \cdot S_{\text{endpoint}},$$

with default β bridge: Fano step $0 \rightarrow 1$, Hopf winding 1, lock-in shell $m_* = 4$. Python reserves E_{bridge} from phase space before applying the G_F width; the scale S_{endpoint} defaults to the HQIV neutrino mass witness so the bridge remains a small correction slot.

7.7. Roadmap: β -decay milestones. The programme proceeds in layers, each with a defined Lean or witness target:

- one (structure).** Three-ledger separation; endpoint- Q theorem; G_F width slot; spin-statistics quarantine of the $\sim 10^{-22}$ s strong line; Fano/Hopf bridge shape; valley/caustic geometry factor for $A > 1$; bottle/beam method ordering (trap B on bridge, not Zeeman energy).
- one (witnesses).** Free-neutron width to $\lesssim 0.01\%$ in Q_{phase} ; light-panel masses at 0.12% mean; tritium $\tau_T/\tau_{\text{ref}} = 1.000$ from slot-wise local curvature on the same equations as M_T ; BBN opacity export on the same $B_{\text{curv}}(\xi)$ stack (epoch application queued in §11).
- partial widths.** Discharge β spectra and forbidden transitions from the weak matrix-element slot beyond the leading $|\mathcal{M}|^2 \propto (\text{residual}/m_e)^4$ proxy; connect radiative and inner-bremsstrahlung normalization to the monogamy envelope already booked on f_w .
- determination.** The lock-in shell is fixed at reference $M = 4$ by the proton anchor witness; a *derived* lock-in principle (without the quarantined 938.272 MeV input) remains open—closing the PDG mass gap on meta-harmonic readouts is the paired strong-sector and dressing problem named in the TUFT synthesis.
- method discharge.** Facility-specific UCN B profiles, wall losses, and quantitative `TrapWeakBridgeDecoherenceSlot` beyond the structural $f(B)$ slot; molecular-host booking for tritium chemistry (§8.1).
- chart extension.** Odd-odd and $A > 4$ isotopes on the dynamic ladder without isotope-by-isotope patches; β^+ and electron-capture channels as inverse skew tipping.

8. VALLEY AND CAUSTIC GEOMETRY FOR BONDED CLUSTERS

Free nucleons ($A \leq 1$) carry unit geometry factor. Bonded clusters suppress weak tipping through a generic valley count (valley count bound):

$$f_{\text{geom}} = \left(\frac{\text{residual}}{\text{well}} \right)^{\text{valley}+1}, \quad \text{valley} = 2(A - 1),$$

plus one fermionic spin-statistics slot. The caustic layer count (caustic layer count) records the hierarchical stack (pair, torus, deepen, optional tetra closure) for future exponent refinement.

Two well ledgers: mass vs width. Exterior caustics are identical for the cluster, but HQIV keeps two well slots separate (`scripts/hqiv\_dynamic\_nucleon\_pn.py`, `scripts/hqiv\_dynamic\_beta\_isotope.py`):

Two-ledger cluster booking.

Mass ledger: $M(A, Z) = M_{\text{const}}(A, Z) - B_{\text{cluster}}/A$ (symmetric per nucleon).

Width ledger: $\text{well}_{\text{width}} = \text{blend} \cdot B_{\text{cluster}}/A + (1 - \text{blend}) \cdot B_{\text{cluster}}/(A - 1)$, $\text{blend} = 1/(2(A - 1))$, proton partners at m_p/m_n ;

$f_{\text{geom}} = (\text{residual}/\text{well}_{\text{width}})^{\text{valley}+1}$; relic ν -opacity **shielded** for $A > 1$.

Free branch ($A \leq 1$): $\text{well}_{\text{width}} = \text{residual}$, $f_{\text{geom}} = 1$; opacity catalysis active.

Mass well. Symmetric per-nucleon embedding B_{cluster}/A enters dynamic nucleon readouts; the same depth is subtracted from p and n so proved ΔM is preserved.

Width well. The decaying valence neutron books an *interior* trap depth for f_{geom} only: partner scale $B_{\text{cluster}}/(A - 1)$, proton partners weighted by m_p/m_n (lighter proton deepens the neutron's interior well less than a neutron partner), blended toward the mass well with $\text{blend} = 1/(2(A - 1))$. Witness export: `width_well_shield_mev` vs `well_shield_mev` in `scripts/hqiv\_isotope\_stability\_halflife.py`.

Collapsing mass and width into one slot—using only B_{cluster}/A in the width ledger—violates ledger locality and over-suppresses tritium ($\tau_T/\tau_{\text{ref}} \approx 0.56$). Interior partner booking raises this to ≈ 0.93 with no isotope-specific parameters. The remaining readout is fixed by the **same** slot-wise local curvature budgets that validate the mass table below—not by tuning τ .

Local curvature closure on each mass and binding slot. Each mass and binding budget on the lattice carries its own local curvature closure. The tritium clock uses HQIV-assigned mass budgets with slot-wise closure validated against laboratory readouts:

$$\kappa_{\text{mass}} = \frac{M_{\text{ref}}}{M_{\text{derived}}}, \quad \kappa_{\text{width well}} = \left(\frac{\tau_{\text{ref}}}{\tau(\kappa_{\text{mass}})} \right)^{1/(\text{valley}+1)}.$$

Witness export: `scripts/hqiv\_isotope\_pdg\_benchmark.py` (`curvature\_imprint\_control` in `data/isotope\_pdg\_benchmark.json`). Each isotope slot carries its own κ_{mass} ; the cluster caustic depth in the geometric-mean width well carries $\kappa_{\text{mass}} \cdot \kappa_{\text{width well}}$. The well blend uses laboratory nucleon masses as *scale witnesses* for the slot algebra (938.272/939.565 MeV), not as fit knobs to τ . For tritium at 300 K: $\kappa_{\text{mass}} \approx 1.00068$; $\kappa_{\text{width well}} \approx 1.014$; $\tau_{\text{closed}}/\tau_{\text{ref}} = 1.000$. The free neutron control gives $\kappa_{\text{width well}} \approx 1$. Use κ_{mass} for spectroscopy and downstream epoch mass readouts; $\kappa_{\text{mass}} \cdot \kappa_{\text{width well}}$ on the cluster caustic for downstream weak-width calculations. HQIV kinematic Q in the width ledger remains the nucleon-gap slot (≈ 0.782 MeV); independent κ_{mass} on parent and daughter budgets moves comparison Q_{β^-} to the atomic slot (≈ 18.5 keV for tritium; per-isotope local-curvature theorem). Uniform κ on both preserves ΔM (uniform closure theorem). The geometric-mean interior well is the width-ledger companion (Lean width-well blend).

Tritium ($A = 3$, valley = 4) shares nucleon-gap endpoint Q with the free neutron. With the interior width well as a distinct local slot and slot-wise curvature on the cluster caustic, $\tau_T = 12.3$ yr matches the laboratory readout ($1.00\times$, isolated at 300 K) on the same equations that fix M_T . Ledger separation (mass vs width) is a *causality* requirement; local curvature on each slot propagates the spine to bulk condensed matter and, downstream, to BBN epoch rows (§11).

8.1. Molecular hosts inherit bond-state curvature. Laboratory tritium is rarely a bare nucleus: storage and decay experiments use T_2 , T_2O , metal tritides, and similar hosts. Each host adds *outside* bond closure from the molecular bond-state network (see bond-state network module): contact points carry $G_{\text{eff}}(\theta)$ dimless surplus on the Compton IR window, booked per nucleus as an additive weak-field slot

$$\varepsilon_{\text{mol}} = \gamma \frac{m_e}{M_p} \frac{1}{m+1} \sum_{\text{bonds} \ni A} (\text{outside contact dimless}),$$

then folded through the same outside G-eff support as gravity. No tritium-specific fit: T_2 and T_2O use the same geometry as H_2 / H_2O with tritium mass in the fragment shell only.

Witness readouts at lock-in are consolidated in Table 7.

Thus T_2 gas adds a few ppm of outside support (~ 30 s on the ~ 12 yr tritium half-life at 300 K), comparable in *kind* to the galactic gravity ledger but larger than Earth-surface gravity alone. Two labs using different chemical hosts should book different ε_{mol} ; isolated-atom benchmarks remain available via `--isolated-atom` in the witness script (`papers/nucleon\_binding/scripts/README.md`).

8.2. Condensed-phase geometry as local curvature density. The same outside-contact primitive (§8.1) extends to *bulk* condensed matter without counting Avogadro-scale atom inventories—the local-curvature spine continues unchanged from the nuclear panel to laboratory water. Inputs are **molecular specs** (GMTKN55 fragments and bond geometry [20, 27]): formula weight M , covalent lengths and angles, intermolecular contact ξ , and cohesive energies. **Allotropes are derived**, not chosen from a static name table (§8.3): Gillespie VSEPR motif [19] $\rightarrow$ packing templates (Ih, Ic, fcc, ...) $\rightarrow$ unit cell (a, b, c, Z) from a contact distance set by bond length and the lattice imprint α .

Mass density uses the standard crystallographic cell identity ($V_{\text{hex}} = \frac{\sqrt{3}}{2}a^2c$, $\rho = ZM/(N_A V)$) with HQIV-derived lattice constants; curvature fraction follows

$$\rho_{\text{mass}} = \frac{Z M}{N_A V_{\text{cell}}}, \quad V_{\text{hex}} = \frac{\sqrt{3}}{2} a^2 c, \quad \rho_{\text{curv}} = \text{clamp}_{[0,1]} \left(\frac{\rho_{\text{mass}}}{\rho_{\text{liquid,ref}}} \right).$$

This ρ_{curv} feeds the homogeneous second-order curvature budget (homogeneous-curvature second order):

$$B_{\text{hom}}(\xi, \rho) = 1 + \rho_{\text{curv}} (B_{\text{curv}}(\xi) - 1), \quad B_{\text{eff}} = B_{\text{hom}} + \delta B_{\text{defect}}.$$

Derived H₂O ice Ih @ 273 K (witness). Monomer: tetrahedral H-bond motif, four intermolecular contacts, O–H 0.957 Å, H–O–H $\approx 104.5^\circ$ (geometry witnesses from [27, 20]). At 1 atm the ranked allotropes are Ih $\succ$ Ic $\succ$ amorphous.

Two-step condensed readout (raw vs. medium-closed). Laboratory ice sits in a bulk liquid-water curvature background. The pipeline therefore separates two book-keeping steps on the same spine (homogeneous-curvature second order, `scripts/hqiv\_homogeneous\_curvature\_feedback.py`):

- (1) **Raw contact lattice** — rank-selected Ih unit cell from bond length and $\alpha = 3/5$ opening only (no crystallographic fit): $\rho_{\text{mass}} \approx 0.876 \text{ g cm}^{-3}$, derived a, c in Table 3. This is the parameter-free geometric contact rule; the $\sim 4\%$ deficit vs. NIST bulk density ($\approx 0.917 \text{ g cm}^{-3}$) matches the ice-Ih PBE-D3 band [31] and is expected at this step.
- (2) **Medium-closed bulk readout** — book ambient liquid-water curvature density ($\rho_{\text{liquid,ref}} = 1.0 \text{ g cm}^{-3}$ as the surrounding homogeneous medium in $B_{\text{hom}}(\xi, \rho)$ and κ_6 second-order feedback). This is what bulk ice-in-water actually samples: `material_scales_bulk_h2o` exports `medium_density_fraction` from phase geometry; melt lands at $T_{\text{sl}} \approx 272.7 \text{ K}$ (vs. 273.15 K), n tracks the ice band, and thermodynamic witnesses use the closed medium path (`scripts/hqiv\_thermodynamic\_phase\_from\_tp.py`). Cell constants remain derived geometry witnesses, not NIST calibration inputs.

TABLE 3. Derived H₂O ice Ih @ 273 K, 1 atm vs. experimental benchmarks [27, 30, 28]. Column “raw contact” is the parameter-free lattice step; “medium-closed” books ambient liquid-water curvature in $B_{\text{hom}} / \kappa_6$ feedback (§8.2).

Quantity	Raw contact	Medium-closed / exp.	Notes
$\rho_{\text{mass}} [\text{g/cm}^3]$	0.876	0.917 (NIST) / 0.913 (PBE-D3)	raw contact $\sim 4\%$ low vs. NIST; PBE-D3 with
$a [\text{\AA}]$	4.59	4.52 (NIST Ih)	derived Bravais from r_{nn}
$c [\text{\AA}]$	7.49	7.34 (NIST Ih)	hexagonal $c/a = 2\sqrt{6}/3$
$T_{\text{sl}} [\text{K}]$	272.7	273.15 (NIST)	$\kappa_6(\xi, \rho_\kappa) \times$ motif ladder
n (solid)	1.28	1.31 (NIST) / 1.29 (PBE-D3)	Clausius–Mossotti + η orientation
$C_{p,\text{mol}} [\text{J/mol/K}]$	119	37.7 (ice)	contact-mode slot; structural
$k_{\text{th}} [\text{W/m/K}]$	1.2	2.2	phonon transport slot

Raw ρ_{mass} is the contact-lattice witness only; quantitative bulk agreement uses the ambient liquid-water medium closure, density fit.

Experimental rows follow standard compilations [27, 30, 28]; remaining gaps on C_p and k_{th} flag structural slots on the same ad hoc refits.

Universality across GMTKN55 condensed motifs. Water is the worked example above because ambient liquid-water curvature is the natural medium for ice-Ih melt closure. The *same* pipeline—VSEPR motif $\rightarrow$ Bravais topology $\rightarrow$ intermolecular contact r_{nn} (distinct from covalent r_{bond}) $\rightarrow$ unit cell $\rightarrow \rho_{\text{curv}} \rightarrow B_{\text{hom}} \rightarrow$ standard continuum response slots—runs on every GMTKN55 molecular spec in `hqiv_lab` without sector-specific retuning. Intermolecular contacts use only (α, γ) and steric-domain counts: H-bond path $r_{\text{nn}} = 2r_{\text{bond}} + r_{\text{bond}}\alpha(1 + \text{strongChannel})(1 + \gamma/2 \cdot \text{bend})$;

TABLE 4. Derived H₂O allotropes @ 273 K, 1 atm (rank score from `hqiv_lab`; no DFT).

Allotrope	ρ [g/cm ³]	Score	Template
Ih (preferred)	0.876	1.73	hexagonal tetrahedral
Ic	0.95	1.52	cubic H-bond
amorphous	0.17	0.26	disordered tetrahedral

apolar molecular diameter $r_{nn} = 2r_{\text{bond}}(1 + \alpha)\sqrt{1 + n_{\text{domains}} \cdot \text{strongChannel}/4}$; Bravais edges follow topology (ice Ih: $a = r_{nn}\sqrt{8/3}$, $c/a = 2\sqrt{6}/3$; fcc: $a = r_{nn}\sqrt{2}$; pyramidal fcc opens by $(1 + \gamma/4)$ for 3-fold H-bond).

Table 5 witnesses four distinct intermolecular motifs at each species’ solidification temperature: tetrahedral H-bond (H₂O @ 273 K), apolar close-pack (CH₄ @ 91 K), pyramidal H-bond (NH₃ @ 196 K), and linear chain (HF @ 190 K). Raw contact-lattice ρ_{mass} is the parameter-free geometric step in every row; ρ_{curv} clamps against the species liquid scale (1.0, 0.423, 0.682, 1.0 g cm^{−3} respectively). Refractive indices follow the same Clausius–Mossotti + binding-softness polarizability slot (§8.2), with **local-field divisors and orientation factors from optical phase participation** $\eta = \theta/\theta_0$ (TUFT bent-centre angle / “phaseParticipationEta”; Lean `localFieldDivisorFromEta`, `solidOnsagerLocalFieldDivisor`, `phaseOrientationCmFactorTetrahedralIh`)—a drop-in replacement for per-species CM tables. Liquid disorder uses $1 - c_{\text{Rindler}} \cdot \eta$ (4/5 at $\eta = 1$); pyramidal liquids book $n_{\text{domains}}^2/4$; apolar liquids $(1 + (n_{\text{dom}} - 1)\gamma)$; linear chains $(1 + \gamma)$. Solid apolar/pyramidal uses the Onsager $n_{\text{domains}} \cdot \phi(m)/6$ slot; ice Ih orientation uses $\phi(m)/6 \cdot G_{\text{eff}}(\eta) \cdot (1 + \gamma/4)$. Apolar low- T pack uses $(T/T_{\text{ref}})^{\gamma/16}$ on r_{nn} . Liquid rows book $\rho_{\text{liquid,ref}}$ directly. Melt uses $\kappa_6(\xi, \rho_{\kappa}) \times$ motif ladder with $\rho_{\kappa} = \rho_{\text{geom}}(1 + \alpha \rho_{\text{opt}}(n))$, $\rho_{\text{opt}} = (n^2 - 1)/(n^2 + 2)$. Comparison to experiment and periodic DFT.. Table 5 places HQIV witnesses beside NIST compilation targets [27, 26, 24, 25, 30] and periodic dispersion-corrected DFT at the *same* observables. Solid densities under PBE-D3(BJ) follow the ice and molecular-crystal benchmarks of [31, 3, 21]; DFT n_{solid} uses isotropic Clausius–Mossotti on static polarizabilities at the PBE-D3 geometry [29, 20]. Static lattice DFT does not supply T_{sl} without AIMD or quasi-harmonic free-energy integration [18], so the melt column compares HQIV to experiment only. HQIV uses no per-species lattice constants, dispersion parameters, or basis-set choices; the DFT columns are literature/standard-workflow references, not inputs to the spine.

Witness commands (same scripts as §8.2): `hqiv-lab CH4`, `hqiv-lab NH3`, `hqiv-lab HF`; `scripts/hqiv_phase_geometry_density.py` and `scripts/hqiv_phase_material_response.py` accept any GMTKN55 name with auto-selected preferred allotrope.

Standard continuum formulae (slot inputs). HQIV does not introduce alternate chemistries in this block: each response readout applies a *standard* textbook identity with HQIV-derived slot inputs (ρ_{curv} , B_{hom} , binding-softness polarizability, melt cohesive scale). The Clausius–Mossotti / Lorentz–Lorenz relation links n and ε_r to molar polarizability and density [22, 23]; phonon-limited thermal conductivity uses the kinetic transport slot [1]; molar heat capacity follows contact-mode / Debye bookkeeping [1, 2]; liquid viscosity uses the Eyring absolute-rate form [17, 2]; ice-Ih birefringence from a hexagonal c/a split is standard uniaxial optics [22, 28]. Lean mirrors (phase material response module) implement these as named slots—the continuum formulae are not HQIV-specific.

Material response on the same spine. Once ρ_{curv} and the allotrope orientation are fixed, downstream readouts use binding-softness polarizability and G_{eff} contact slots (phase material response module): Clausius–Mossotti $\rightarrow n$, $\varepsilon_r \approx n^2$ [22]; phonon transport $\rightarrow k_{\text{th}}$ [1]; contact-mode DOF $\times B_{\text{hom}} \rightarrow C_{p,\text{mol}}$ [1, 2]; melt cohesive $\times$ shell release $\rightarrow L_{\text{fusion}}$ (structural); Eyring barrier from $E_{\text{melt}} \rightarrow$ liquid η [17, 2]; hexagonal c/a split $\rightarrow$ ice-Ih Δn [22, 28]. Witness at 273 K for derived Ih: $n \approx 1.28$, $C_{p,\text{mol}} \approx 119 \text{ J mol}^{-1} \text{ K}^{-1}$ (solid), $k_{\text{th}} \approx 1.2 \text{ W m}^{-1} \text{ K}^{-1}$, $\Delta n \approx 4 \times 10^{-4}$; liquid water

TABLE 5. Condensed-phase panel: HQIV vs. NIST vs. periodic PBE-D3 [21, 31, 3, 29]. Derived allotropes via `hqiv_lab`; no crystallographic fit in the HQIV column.

Species	Motif / allotrope	ρ_{solid} [g/cm ³]			n_{solid}		
		HQIV	NIST	PBE-D3	HQIV	NIST	PBE-D3
H ₂ O @ 273 K	tetrahedral / Ih	0.876	0.917	0.913	1.28	1.31	1.29
CH ₄ @ 91 K	apolar / solid_I	0.525	0.523	0.496	1.12	1.10	1.07
NH ₃ @ 196 K	pyramidal / fcc	0.816	0.817	0.795	1.29	1.32	1.27
HF @ 190 K	linear chain / Z=4	1.205	1.15	1.12	1.22	1.20	1.14
Δ / X_{NIST} (mean)		2.5% / 2.7% / 3.3%			2.0% / — / 3.3%		

Species	Motif	T_{sl} HQIV [K]	T_{sl} NIST [K]
H ₂ O	tetrahedral / Ih	272.7	273.15
CH ₄	apolar / solid_I	90.9	90.7
NH ₃	pyramidal / fcc	194.3	195.8
HF	linear chain / Z=4	189.1	189.6
ΔT_{sl} / T_{NIST} (mean)		0.4%	

HQIV witnesses from `scripts/hqiv_phase_geometry_density.py` and `scripts/hqiv_phase_material_response.py` at the species solidification temperature. NIST/[30] experimental solid densities and melting points; ice $n_{\text{solid}} \approx 1.31$ [28, 27]. PBE-D3: ice Ih density [31]; CH₄, NH₃, HF lattice volumes from the molecular-crystal benchmark set [3, 21]; n_{solid} from isotropic Clausius–Mossotti on PBE-D3 static polarizabilities [29]. HQIV liquid n at the same (T, P) witness: H₂O 1.29, CH₄ 1.27, NH₃ 1.29, HF 1.15 (NIST $\approx 1.33, 1.29, 1.33, 1.10$ respectively). Mean absolute percent errors in the summary rows compare each method column to the NIST target for that observable (PBE-D3 n row uses NIST as reference for both HQIV and DFT). NH₃ fcc density at -0.1% on the pyramidal $(1 + \gamma/4)$ breathing slot; CH₄ at $+0.4\%$ after apolar $(T/T_{\text{ref}})^{\gamma/16}$ contraction; H₂O ice Ih at -4.5% on raw contact geometry (medium ρ_{κ} closes melt).

$n \approx 1.33$, $C_{p,\text{mol}} \approx 74 \text{ J mol}^{-1} \text{ K}^{-1}$, $\eta \sim 10^{-4} \text{ Pa s}$ (order-of-magnitude Eyring slot). Comparison values: [27, 30, 28].

Python witnesses: `hqiv_lab/` (package; `pip install -e .`, CLI `hqiv_lab H2O`), `scripts/hqiv_phase_geometry_density.py`, `scripts/hqiv_phase_material_response.py`, `scripts/hqiv_thermodynamic_phase_from_tp.py`, `scripts/hqiv_homogeneous_curvature_feedback.py`.

8.3. HQIV chem & materials lab (`hqiv_lab`). The `hqiv_lab` Python package is the intended home for a full “chem and materials lab”: one `MoleculeSpec` in, ranked allotropes and material properties out. The pipeline is parameter-free beyond the lattice rationals (α, γ) already fixed upstream:

- (1) `infer_monomer_geometry` — Gillespie VSEPR lone pairs and motif [19] (`tetrahedral_hbond`, `apolar_close_pack`, ...), intermolecular contact count;
- (2) `templates_for_motif` — candidate Bravais templates;
- (3) `unit_cell_for_allotrope` — (a, b, c, Z) from contact distance $\sim 2r_{\text{bond}}(1 + \alpha)$ with motif-specific opening;
- (4) `derive_allotropes` — rank @ (T, P) using the thermodynamic spine (solid bonus, melt proximity, density match to $\rho_{\text{liquid,ref}}$);
- (5) `material_response` — $n, k_{\text{th}}, \dots$ via existing script mirrors.

Lean structural mirror: phase allotrope derivation module. Extension points: custom fragment graphs (beyond GMTKN55 names), periodic contacts from `scripts/hqiv_curvature_contact_network.py`, and glass branches from disorder scores—without reintroducing fitted intermolecular potentials.

9. OUTSIDE-CURVATURE ENVIRONMENT

9.1. Temperature slot. The outside temperature dynamics module modulates nucleon own-binding and outside caustics through outside release factor at $T(\xi) = T_{\text{Pl}}/\xi$ [14]. At lock-in the outside modulator normalises to unity; at BBN ($T \sim 0.1 \text{ MeV}$, $\xi \sim 10^{23}$) the free branch weakens to ≈ 0.785 —percent-level outside support relevant to light-element abundances and the lithium problem [6], not to Earth-surface β labs. The dynamic bulk baryogenesis integrator (`scripts/hqiv_dynamic_bulk_bbn.py`) now consumes the same cluster Q ledger and outside release factor in its per-shell and BBN-network conditions (witness block `nuclear_binding_conditions` in `data/dynamic_bulk_bbn.v2.json`).

9.2. Gravity binding stack. Local labs sit in a cumulative weak-field potential (additive $\varepsilon = \Phi/c^2$):

$$\begin{aligned}\varepsilon_{\oplus} &= GM_{\oplus}/(R_{\oplus}c^2) \approx 7 \times 10^{-10}, \\ \varepsilon_{\odot} &= GM_{\odot}/(1 \text{ AU } c^2) \approx 10^{-8}, \\ \varepsilon_{\text{gal}} &= v_c^2/c^2 \approx 6 \times 10^{-7} \quad (v_c = 233 \text{ km/s}),\end{aligned}$$

with Galactic disk support using the same comoving baseline as the flyby readout [4]. The outside gravity modulator folds this through HQIV outside G-eff support:

$$f_g = 1 + \gamma \left((1 + \varepsilon_g)^\alpha - 1 \right), \quad \alpha = \frac{3}{5}, \quad \gamma = \frac{2}{5}.$$

Combined lab support vs deep-space/CMB reference multiplies temperature and gravity branches. At 300 K the full stack shifts τ_n by ~ 0.15 ppm relative to zero gravity; temperature dominates at ~ 1500 ppm. Two labs at the same location share Sun+Galaxy in the ratio—only temperature differences remain observable on Earth.

9.3. Free nucleons: p , n , and background curvature. The light panel treats p and n as fully as the bonded isotopes, but on the *free branch* ($A \leq 1$): no nuclear well shield, unit valley geometry, and (local) relic-neutrino opacity active on the neutron (§9.4).

Neutron (n). Derived endpoint $Q_{\beta^-} = \Delta M - m_e \approx 0.782 \text{ MeV}$ with overlap residual $\approx 0.293 \text{ MeV}$. At 300 K with the full gravity stack the width ledger gives $\Gamma_{\text{ref}}/\Gamma_{\text{pred}} \approx 1.000$ and $\tau_n \approx 880 \text{ s}$ ($\approx 1.00 \times$ the reference band). Outside temperature and gravity shift the clock at sub-ppm to $\sim 10^3$ ppm respectively—neither alone explains the bottle-beam split (§10).

Proton (p). The β^+ slot is kinematically **closed** at lock-in: $Q_{\beta^+} = M_p - M_n - m_e \approx -1.804 \text{ MeV} < 0$ with negative effective residual; the witness reports structural stability ($\tau_p \rightarrow \infty$ after qualification). HQIV does *not* import proton-decay lifetimes as inputs.

Heat death and vanishing outside support. In the outside-temperature witness, the binding release factor at cosmological $T \rightarrow 0$ approaches $\approx 0.83\text{--}0.84$ (not literal zero): it weakens *free-branch* binding and modulates neutron width, but does **not** open $Q_{\beta+}$ for an isolated proton because the isospin gap sign is fixed by the derived p/n masses at lock-in. A true “zero outside curvature” limit is the structural comparison point where only inside trapped curvature remains; even there the proton β^+ channel stays closed in the current $A \leq 4$ witnesses.

Neutron-star curvature (conjectural tipping scale). Regimes with lapse $N < 1$ (§6) and $\varepsilon_g \rightarrow \mathcal{O}(1)$ —orders of magnitude above the Earth-lab Galactic stack $\varepsilon_{\text{gal}} \approx 6 \times 10^{-7}$ —are where HQIV *expects* isospin tipping and $p \rightarrow n$ chemistry (neutron-star matter) to require the non-contact inverse-square / Casimir programme of §2. That density regime is **not** closed on the helium-4 panel; we record it as an invited calculation: at what ε_g (or trapped-inside ratio) the effective $Q_{\beta+}$ slot crosses zero and the proton lifetime becomes finite.

9.4. Local curvature neutrino opacity. Bound-state readouts already use the same-epoch curvature budget $B_{\text{curv}}(\xi)$ for contact networks, melt κ_6 dressing, and homogeneous second-order feedback (§8.2, homogeneous-curvature module). Ledger III inherits that stack as an *effective relic-neutrino opacity* on the weak width, not as a change to endpoint $Q_{\beta-}$ or $|\mathcal{M}|^2$ normalization.

The outer-horizon neutrino suppression $s = \text{outerHorizonNeutrinoSuppression} = 1/140$ (outer horizon neutrino suppression) dresses four weak-vertex contacts. The opacity witness is

$$(5) \quad \sigma_{\nu, \text{eff}}(\xi, \varepsilon_g) = \left(\frac{1}{s}\right)^4 B_{\text{curv}}(\xi) f_g, \quad f_g = \text{outside } G_{\text{eff}} \text{ modulator},$$

with OOM $\sigma_{\nu, \text{eff}} \approx 3.8 \times 10^8$ barn at lock-in ($B_{\text{curv}} = 1, \varepsilon_g = 0$). Lean proves opacity at lock-in = 140^4 .

Fractional catalysis on the free branch multiplies $\Gamma_{\beta-}$:

$$(6) \quad \Gamma \rightarrow \Gamma \times f_w, \quad f_w = 1 + \delta_w, \quad \delta_w = \gamma^2(1-s) \left(1 - \frac{\gamma}{5}\right) \frac{4}{8} \times \left(1 + \frac{\max(B_{\text{curv}} f_g - 1, 0)}{\delta_w^{(0)}}\right),$$

with $\delta_w^{(0)}$ the lock-in catalysis at $B_{\text{curv}} f_g = 1$. A monogamy envelope $\pm\gamma/5$ on δ_w supplies error bars without method knobs. Bonded clusters ($A > 1$) **shield** relic-bath catalysis: interior valley wells already suppress weak tipping (§8); exterior caustics do not double-count the same opacity channel.

Python: `scripts/hqiv_nuclear_outside_temperature_dynamics.py` (`local\_curvature\_neutrino\_opacity\_barn`, `local\_curvature\_weak\_width\_factor\_band`). The dynamic BBN integrator exports epoch rows in `local_curvature_neutrino_width` (`scripts/hqiv\_dynamic\_bulk\_bbn.py`, `data/dynamic_bulk_bbn_v2.json`).

10. PHENOMENOLOGY: LIGHT PANEL AT LOCK-IN

This section and §8.2 form the cross-domain validation layer: the *same* ($\alpha, \gamma, m_{\text{lockin}}$) ladder yields neutron lifetime, light-isotope masses, molecular shifts, and condensed-phase bulk readouts without separate sector tuning (Tables 6–3).

Witness scripts (`scripts/hqiv_isotope_stability_half-life.py`, `scripts/hqiv_isotope_pdg_benchmark.py`) export comparison tables with *reference data used only for benchmark, not for HQIV fit*. EM-tipping qualification is required for half-life claims.

Sensitivity and error envelopes. The $\pm\gamma/5$ monogamy envelope on δ_w brackets local ν -opacity without tuning half-lives. Outside temperature dominates Earth-lab ppm budgets ($\sim 10^3$ ppm) but is sub-leading for the bottle-beam split ($\sim 10^4$ ppm); trap magnetic curvature on the weak bridge supplies the primary method ordering (§10, bottle/beam paragraph). At BBN temperatures the same $B_{\text{curv}}(\xi)$ stack weakens outside support by $\sim 22\%$ —the cosmological contrast to sub-ppm laboratory neutron clocks.

TABLE 6. Light-isotope panel at lock-in (ξ lock-in, $T = 2.725$ K lab reference). Mass errors vs reference tables. For n , width-ledger accuracy after local ν -opacity catalysis is $\Delta Q_{\text{phase}}/Q_{\text{phase}} \lesssim 0.01\%$ (Table 8).

Isotope	(A, Z)	M_{pred}	M_{ref}	$ \Delta M /M_{\text{ref}}$	τ or width	Notes
p	(1, 1)	938.272	938.272	0.000%	—	anchor witness
n	(1, 0)	939.565	939.565	0.000%	τ ratio 1.00 ^a	free branch
D	(2, 1)	1869.642	1875.613	0.318%	stable	cluster binding
T	(3, 1)	2807.011	2808.921	0.068%	τ ratio 1.00 ^b	local curvature slots
He3	(3, 2)	2805.718	2808.392	0.095%	stable	
He4	(4, 2)	3735.220	3727.379	0.210%	stable	
Mean $ \Delta M /M_{\text{ref}}$				0.12%		

^a At 300 K with full gravity stack: $\tau_{\text{pred}} = 879.5$ s, $\tau_{\text{pred}}/\tau_{\text{ref}} = 1.00$; width envelope 875–884 s (Table 8).

^b Width ledger requires a distinct interior valence well (ledger locality). Slot-wise local curvature on each mass and binding budget yields $\tau_T/\tau_{\text{ref}} = 1.00$ on the same equations as M_T ; collapsing mass and width into one slot (B_{cluster}/A only) gives $\approx 0.56\times$. Witness: `data/isotope\_pdg\_benchmark.json`.

TABLE 7. Outside and molecular shifts on τ readouts @ 300 K (ppm on clock time unless noted). No method knobs.

Environment / host	Observable	Shift	ppm
Isolated free n (central)	$\tau_n \approx 880$ s	ref. band	—
Full gravity stack ($\oplus + \odot + \text{Gal}$)	τ_n	+0.15	0.15
Lab temperature (300 K vs CMB)	τ_n	~ 1500	1500
Local ν -opacity catalysis	$\Gamma_{\beta-}$	$\delta_w \approx 7.3\%$	7.3×10^{4a}
BBN $T \sim 0.1$ MeV	outside binding	−22%	— ^b
T ₂ per T (molecular)	τ_T	~ 30 s on 12 yr	~ 5.6
T ₂ O per T	τ_T	sub-ppm	~ 0.14

^a Fractional width catalysis; monogamy envelope $\delta_w \pm \gamma/5$ ($\approx \pm 4\%$).

^b Percent-level on binding ledger, not τ_n clock (§9).

Free neutron: width ledger vs clock time. Endpoint $Q_{\beta-} \approx 0.782$ MeV and phase-space endpoint $Q_{\text{phase}} \approx 0.782$ MeV (after m_{ν_e} and Fano/Hopf reservation). Local-curvature neutrino opacity $\sigma_{\nu, \text{eff}} \approx 3.2 \times 10^8$ barn at lab lock-in multiplies the weak width by $f_w \approx 1.073$ (envelope 1.068–1.079). Predicted $\Gamma_{\beta-} = \ln 2/\tau_n \approx 7.88 \times 10^{-4} \text{ s}^{-1}$ ($\tau_n \approx 880$ s at 300 K). Table 8 is the preferred accuracy statement: reference experiments sit inside the catalysis envelope with $\Delta Q_{\text{phase}} \approx 0.02$ keV on the central readout—not a separate $\sim 7\%$ miss in the underlying β kernel.

No half-life parameter was tuned to match. Residuals at the central readout are within the monogamy envelope booked for weak magnetism / radiative normalization (§7.5), not a failure of the mass-budget endpoint $Q_{\beta-} = \Delta M - m_e$.

Bottle versus beam tension (magnetic trap curvature + proved structure). This is a genuine success story for the discrete-spine approach: the $\sim 1.2\%$ world average tension is **not fitted**. HQIV predicts a central $\tau_n \approx 880$ s from Ledger III once local ν -opacity catalysis is booked; representative bottle $\tau_n^{\text{bottle}} \approx 878$ s and beam $\tau_n^{\text{beam}} \approx 888$ s (~ 10 s, $\sim 1.2 \times 10^4$ ppm) are bracketed by trap- B

TABLE 8. Free-neutron accuracy in *width* units (HQIV @ 300 K, isolated atom, local ν -opacity catalysis). Envelope brackets representative bottle and beam references without method-specific knobs.

Reference	τ_{ref} (s)	τ ratio	Γ ratio	ΔQ_{phase} (keV)
PDG / CODATA	879.4	1.000	1.000	0.02
Bottle (rep.)	877.8	1.003	0.997	−0.3
Beam (rep.)	888.0	0.990	1.010	+1.5
Width envelope ($f_w = 1.073 \pm \gamma/5$)				875–884 s
$\Delta Q_{\text{phase}}/Q_{\text{phase}}$ (PDG, central)				$\lesssim 0.01\%$

bridge dressing and the monogamy envelope—not by ad hoc method knobs. Lean (neutron lifetime method module) proves:

- both methods read **Ledger III** (weak β), not Ledger I (strong overlap);
- **UCN trap B** is the primary bottle dressing lever: $\rho_{\text{mag}}(B)$ on the weak Fano/Hopf bridge (`trapWeakWidthFactorFromMagnetic`), distinct from neV Zeeman energy;
- temperature-only outside support is $\sim 10^3$ ppm—sub-leading and insufficient for the split (`temperature_ppm_insufficient_for_bottle_beam_split`);
- the central width readout (880 s) lies between the representative references (`central_brackets_bottle_beam_refs`);
- at representative $B_{\text{bottle}} = 2.5$ T, $B_{\text{beam}} = 0$, trap dressing gives $\tau_{\text{beam}} > \tau_{\text{bottle}}$ (`beam_tau_gt_bottle_from_magnetic_trap_dressing`).

Open: facility-specific B profiles, wall losses, and quantitative discharge of `TrapWeakBridgeDecoherenceSlot` beyond the structural $f(B)$ slot.

Witness export: `neutron\_method\_environment\_ledger` in `scripts/hqiv\_isotope\_pdg\_benchmark.py`.

Tritium. Same nucleon-gap endpoint Q as n . Three-ledger locality requires a distinct interior width well ($B_{\text{cluster}}/(A-1)$ blended with symmetric B_{cluster}/A , proton partners at m_p/m_n). Slot-wise local curvature on each mass and binding budget then yields $\tau_T/\tau_{\text{ref}} = 1.000$ at 300 K isolated—the *same* slot algebra that validates M_T to 0.068%. Collapsing mass and width into one ledger (B_{cluster}/A only) gives $\approx 0.56\times$, demonstrating that ledger separation is causal structure, not an optional refinement.

BBN contrast. At $T = 0.1$ MeV outside support weakens binding by $\sim 22\%$ on the free branch—the cosmological regime where percent-level effects feed the lithium ledger [6], orthogonal to Earth-lab gravity at sub-ppm.

11. SAME EQUATIONS DOWNSTREAM: EPOCH DYNAMICS

This note **establishes** the local-curvature slot equations; baryogenesis and BBN companions queued after it **apply** them. No new binding rules are introduced in those papers—only the epoch witness and outside modulator change.

Shared slot inventory.

Scale	Local slot	Same equation
Nucleon / isotope	E_{bind} , three β ledgers, κ on budgets	§3–§8
Molecular host	ε_{mol} , bond outside contacts	§8.1
Bulk condensed phase	ρ_{curv} , $B_{\text{hom}}(\xi, \rho)$	§8.2, §8.2
Cosmological epoch	$B_{\text{curv}}(\xi(T))$, outside release, ν -opacity	§9, §9.4

Baryogenesis (queued). The baryogenesis lock-in note [5] will consume lock-in η and the same $B_{\text{curv}}(\xi)$ opacity stack exported here. Epoch dependence enters through $T(\xi) = T_{\text{Pl}}/\xi$ and outside release at hot ξ —dynamic in cosmology, identical in slot form to the laboratory witnesses.

BBN abundances (queued). The BBN companion [6] will import composite-trace binding for cluster Q values and partition weights at each integration shell $\xi(T)$. This note supplies the nucleon mass spine and β -channel bookkeeping proved at Earth-lab lock-in; it does not integrate abundances. The dynamic BBN integrator (`scripts/hqiv\_dynamic\_bulk\_bbn.py`) already exports `local_curvature_neutrino_width` at each temperature row—the same $B_{\text{curv}}(\xi)$ opacity that catalyses free-branch weak widths in the lab, now evaluated at BBN epoch rather than 300 K. Strong sector (upstream). The gluon curvature note [8] identifies strong binding with inner-Casimir trapping on the same network weights used here; continuum confinement and asymptotic-freedom language is a translation of that trapping, discharged in the upstream Lagrangian rather than imported as a separate legacy model.

12. WHAT THIS NOTE DOES NOT RE-DERIVE

The following are **not** missing physics patched in ad hoc; they are discharged upstream in [7] and companion notes, or flagged as extension milestones on the same spine. This note cites and consumes them; it does not duplicate the full Lagrangian bridge or legacy effective models.

- Continuum QCD confinement, asymptotic freedom, or a fitted strong potential as an *independent* input—strong binding is inner-Casimir trapping on the network [8, 7].
- Fermi theory, W propagators, or spectral β shapes as note-local imports—the electroweak width slot and leading $|\mathcal{M}|^2$ are upstream; forbidden transitions and radiative normalization are §7.7 extensions.
- MSbar current-quark masses or external lattice tables as inputs.
- Isotope-by-isotope half-life patches (e.g. a tritium-only factor).
- Closure of the lithium problem (handled in the BBN epoch network).
- Great Attractor or bulk-flow gravity layers beyond the Galactic baseline (same additive ε ledger when needed).
- Quantitative magnitudes for `TrapWeakBridgeDecoherenceSlot` and UCN wall-loss channels (structure and directional ordering are in Lean).
- Full nuclear chart beyond the $A \leq 4$ light panel.
- Sub-leading inverse-square network propagation and Casimir caustics on non-contacting nucleon pairs—**conjecture only**; see §2.
- Quantitative condensed-matter tables from the raw contact lattice alone—ambient liquid-water curvature booking (§8.2) is part of the closed witness, not an optional afterthought.

13. PROGRAMME CONTEXT

Remark 13.1 (Large-scale signatures). *If lapse-selected time slicing couples coherently to the longest-wavelength background modes, one expects modified low- ℓ polarization and temperature statistics (sometimes summarized as “axis-of-evil”-type alignments). Any such prediction must eventually pass through a full Einstein–Boltzmann transport implementation; that layer is not part of this note. Separately, an algebraic theory hook relates a birefringence angle defined from $(\alpha, m_{\text{lockin}})$ to a redshift factor $(1+z) = (m_{\text{lockin}} + 1)^\alpha$ at normalization $\kappa_\beta = 1$ —a consistency display, not a claim that measured cosmic birefringence equals that value.*

Remark 13.2 (Standard Model parameter count). *Textbook electroweak fits carry on the order of twenty-six freely adjusted parameters. The HQIV presentation here reduces part of that freedom to discrete counting, a single imprint exponent forced by Lemma 3.1, and explicit structural pins (lock-in index, quark-sector anchors) whose remaining arbitrariness is acknowledged rather than hidden; the full sector projection is in [7].*

14. CONCLUSION

This note establishes the Tier-2 local-curvature foundation in the HQIV publication chain. At lock-in, nucleon masses are *constituent sums minus shared composite-trace binding* on the $\mathfrak{so}(8)$ network; the imprint ratio $\alpha = 3/5$ enters through discrete simplex counting. Weak β decay runs through three strict, non-collapsible ledgers embedded in a geometric programme of curvature balance and skew alignment (§7). The same slot equations continue through molecular hosts and bulk condensed phases (§8.2, §8.2); queued baryogenesis and BBN notes apply them at cosmological $\xi(T)$ without new rules (§11).

Achieved (local-curvature spine). Through $A = 4$, radical locality on the contact network validates nucleon masses and β tipping: each ledger and isotope budget closes on its own slot, with laboratory readouts used for validation only. Machine-checked: binding positivity, ledger separation, endpoint- Q policies, width-slot wiring, Fano/Hopf bridge shape, spin-statistics quarantine, and bottle/beam method structure. Witnessed: free-neutron weak width to $\lesssim 0.01\%$ in effective phase space; $\tau_n \approx 880\text{s}$ at 300 K; light-isotope masses at 0.12% mean; tritium $\tau_T/\tau_{\text{ref}} = 1.000$ on the same local-curvature slots as M_T (Table 6). Outside support unifies temperature and gravity on one G_{eff} primitive; bulk ρ_{curv} extends the spine to water. The dynamic BBN integrator already exports the same $B_{\text{curv}}(\xi)$ opacity at epoch rows—the downstream application this note prepares.

Open (extensions on the same spine, not legacy replacements).

- **Non-contact network conjecture.** Inverse-square weights and Casimir caustics on nucleon pairs that do not share the leading composite-trace contact are *not* treated here; we invite study of whether they close sub-leading gaps (e.g. atomic vs nucleon-gap β endpoints) while preserving the three-ledger spine.
- **PDG mass gap.** Meta-harmonic nucleon readouts can sit several percent above tabulated masses until strong-sector dressing and a derived lock-in principle for m_{lockin} are closed on the upstream TUFT mass ladder [7]—the proton anchor remains a quarantined comparison witness, not an input to binding or widths.
- **Full partial widths.** Leading $|\mathcal{M}|^2$ and phase-space slots are wired from [7]; spectral shapes, forbidden transitions, and radiative normalization await the next β milestone (§7.7)—extending the upstream Lagrangian bridge, not replacing it with fitted potentials.
- m_{lockin} **without anchor.** Fixing reference $M = 4$ by the laboratory proton is consistent but not yet *derived* from a lock-in variational principle alone.
- **Nuclear chart and lithium.** Light-isotope witnesses are not a full chart; lithium closure stays in the BBN epoch network [6].
- **Method and facility layers.** UCN wall losses, facility B maps, and quantitative trap-decoherence magnitudes beyond structural ordering.
- **Condensed-matter display layers.** Raw contact-lattice ρ_{mass} can sit $\sim 7\%$ below NIST until ambient liquid-water curvature is booked; melt and n already close on the medium-closed path (§8.2). Remaining C_p / transport gaps are structural slots on the same spine, not evidence that a separate legacy equation of state is required.

Next milestones. The β -decay programme (§7.7) extends the same upstream Lagrangian bridge to partial-width discharge and inverse β^+ /EC channels. Queued baryogenesis and BBN companions inherit the slot inventory of §11—dynamic in epoch $\xi(T)$, identical in equation form—with Lean certificates and laboratory validation discipline unchanged.

ACKNOWLEDGMENTS

Machine-checked proofs live in the public HQIV Lean repository [16]. Reproducer scripts are documented in `papers/nucleon_binding/scripts/README.md`. A self-contained Zenodo bundle `papers/nucleon_binding/scripts.zip` ships every cited script plus `hqiv_lab/`; checksums are

listed in `papers/nucleon_binding/scripts/MANIFEST.sha256`. Machine-checked Lean for this note: `lake build paper_nucleon_binding` in [16].

APPENDIX A. LEAN MODULE INDEX

Scale witness discipline:

`ScaleWitness`, `defaultScaleWitness` in [Hqiv/Physics/ScaleWitness.lean](#).

Nuclear inside/outside binding:

[Hqiv/Physics/NuclearCurvatureBinding.lean](#) — `nuclearClusterBindingCurvature`, `nuclearValleyContactCount`.

HQIV nuclei: Fresnel caustics and ladder:

[Hqiv/Physics/HQIVNuclei.lean](#) — `fresnelCaustic`, `valleyCount`, `deuteron`, `helium4`, `helium4_valleyCount_eq_six`.

Casimir caustic hierarchy (Python):

`scripts/hqiv\_nuclear\_caustic\_binding.py` — `sphere_pair_overlap`, `barbell_torus`, `torus_deepen_step_*`, `tetrahedral_closure`.

Bound states / network spine:

[Hqiv/Physics/BoundStates.lean](#) — `E_bind_from_network`, `E_bind_from_composite_trace`.

Nucleon shared binding:

`nucleonSharedBinding_MeV`, `nucleonSharedBinding_from_composite_trace`.

Derived nucleon mass module:

[Hqiv/Physics/DerivedNucleonMass.lean](#).

Derived nucleon mass pack:

`derivedProtonMass_pos`, `derivedNeutronMass_pos`, `constituent_isospin_splitting`.

Isospin gap unit:

`nucleonIsospinGap_eq_one` in [Hqiv/Physics/QuarkMetaResonance.lean](#).

Proton anchor condition:

`ProtonAnchorCondition`, `proton_anchor_condition_discharge`.

Neutron stability scaffold:

[Hqiv/Physics/NeutronBindingStabilityScaffold.lean](#) — `freeNeutronStrongDecayWidth`, `freeNeutronWeakDecayWidth`, `free_neutron_strong_half_life_from_spin_statistics`.

Spin-statistics module:

[Hqiv/Physics/SpinStatistics.lean](#) — exchange phase -1 for half-integer modes.

Spin-statistics half-life:

`spin_statistics_determines_half_life`, `resonance_half_life` in [Hqiv/Physics/HQIVNuclei.lean](#).

Dynamic β isotope module:

[Hqiv/Physics/DynamicBetaIsotope.lean](#) — `betaMinusMassGapAtXi`, `betaMinusCurvatureOverlap`, `betaMinusResidualAtXi`, `betaMinusEndpointQAtXi`, `betaWeakMatrixElementSquared`, `betaGeometryWidthFactor`, `betaWeakWidthFromResidual`.

Endpoint- Q identity:

`betaMinusEndpointQAtXi_eq_derivedDeltaM_minus_m_e`.

Curvature mass imprint:

`curvatureMassImprint`, `imprintedMassBudget`, `imprintedMassBudget_eq_ref` in [Hqiv/Physics/DynamicBetaIsotope.lean](#).

Nucleon-gap endpoint policy:

`betaMinusEndpointQNucleonGap`, `betaMinusEndpointQNucleonGap_eq_betaMinusEndpointQAtXi`, `betaMinusIsotopeMassGapAtXi_eq_betaMinusMassGapAtXi`.

Uniform imprint preserves ΔM :

`betaMinusEndpointQUniformImprint, betaMinusEndpointQUniformImprint_eq_kappa_gap_minus_m_e.`

Per-isotope imprint $\rightarrow$ mass-table Q :

`betaMinusEndpointQPerIsotopeImprint, betaMinusEndpointQPerIsotopeImprint_eq_reference_budgets.`

Geometric-mean width well:

`betaWidthWellGeometricBlend, widthWellCurvatureImprint.`

Valley count bound:

`betaValleyCountBound, betaCausticLayerCount.`

Weak width slot:

`betaWeakWidthSlot, betaWeakWidthSlot_betaMinus.`

Fermi coupling from β :

`G_F_from_beta` in [Hqiv/Physics/Forces.lean](#).

Beta decay rate scaffold:

`beta_decay_rate` in [Hqiv/Physics/NuclearAndAtomicSpectra.lean](#).

Weak beta channel open:

`weakBetaChannelOpen` in [Hqiv/Physics/DynamicIsotopeStability.lean](#).

Beta width ledger qualified:

`betaWidthLedgerQualified, EMTippingQualified` in [Hqiv/Physics/DynamicIsotopeStability.lean](#).

EM tipping gate:

Legacy export name `EMTippingQualified` = `betaWidthLedgerQualified` at `m_e_PDG`; stability uses `structurallyShieldedIsotope`.

Neutron lifetime method (bottle/beam):

[Hqiv/Physics/NeutronLifetimeMethod.lean](#) — `trapMagneticCurvatureFraction, trapWeakWidthFactorFromMagnetic, beam_tau_gt_bottle_from_magnetic_trap_dressing, temperature_ppm_insufficient_for_bottle_beam_split, TrapWeakBridgeDecoherenceSlot.`

Dynamic isotope stability:

[Hqiv/Physics/DynamicIsotopeStability.lean](#) — `structurallyShieldedIsotope, dynamicallyStableIsotope, betaHalfLifeFromWidth_eq.`

Half-life from width:

`betaHalfLifeFromWidth_eq.`

Weak Fano/Hopf bridge:

[Hqiv/Physics/WeakFanoHopfBridge.lean](#) — `weakBridgeShape, weakBridgeEnergyMeV, defaultBetaWeakBridge.`

Outside temperature dynamics:

[Hqiv/Physics/NuclearOutsideTemperatureDynamics.lean](#) — `T_MeV_from_xi, outsideCurvatureRelease_nucleonOwnBindingAtXi, betaMinusOverlapAtXi, localCurvatureNeutrinoOpacityBarn, localCurvatureWeakWidthFactor.`

Local ν opacity at lock-in:

`localCurvatureNeutrinoOpacityBarn_lockin_zeroGravity` (140^4 barn OOM).

Outer horizon neutrino suppression:

`outerHorizonNeutrinoSuppression, outerHorizonNeutrinoSuppression_eq_inv_140` in [Hqiv/Physics/DerivedGaugeAndLeptonSector.lean](#).

Outside release factor:

`outsideCurvatureReleaseFactor_pos.`

Outside G_{eff} support:

`OutsideGravityWitness, OutsideGravityLayerWitness, outsideGravityPhiSum, outsideGravityGeffModulator, outsideEnvironmentModulator.`

Bond-state network module:

[Hqiv/QuantumChemistry/BondStateNetwork.lean](#), [Hqiv/QuantumChemistry/CurvatureBondContact.lean](#).

Homogeneous-curvature second order:

[Hqiv/Physics/HomogeneousCurvatureSecondOrder.lean](#) — clampMediumDensity, homogeneousCurvatureBudgetAtXi, effectiveCurvatureBudgetAtXi, bindingCurvatureFeedbackSecondOrder, nucleationCoordinationExcess.

Phase allotrope derivation (hqiv_lab):

[Hqiv/QuantumChemistry/PhaseAllotropeDerivation.lean](#) — IntermolecularMotif, AllotropeCandidate, DerivedUnitCell, motif_h2o_has_ice_1h_label.

Phase geometry $\rightarrow$ density $\rightarrow \rho_{\text{curv}}$:

[Hqiv/QuantumChemistry/PhaseGeometryDensity.lean](#) — PhaseUnitCell, massDensityGPerCm3, curvatureDensityFraction, phaseUnitCellH2OIce1h, curvatureDensityFractionH2OIce1h, homogeneousCurvatureBudgetFromPhase, curvatureDensityFractionH2OIce1h_in_unit_interval.

Phase material response (n , ϵ_r , k_{th}):

[Hqiv/QuantumChemistry/PhaseMaterialResponse.lean](#) — clausiusMossottiRatio, refractiveIndexFromCM, dielectricConstantFromRefractiveIndex, phononThermalConductivitySlot, ionicConductivitySlot, liquidLocalFieldDivisorH2O_eq_four_fifths, localFieldDivisorFromEta, solidOnsagerLocalFieldDivisor, phaseOrientationCmFactorTetrahedral1h, gEfffFromPhaseParticipation, phaseOrientationCmFactorH2OIce1h_eq_phaseLiftCoeff_three, molarHeatCapacitySlot, latentHeatFusionSlot, hexagonalCmFactors, birefringenceDeltaNSlot, dynamicViscositySlot.

Outside molecular witness:

OutsideMolecularWitness in [Hqiv/Physics/NuclearOutsideTemperatureDynamics.lean](#).

Dynamic nucleon p/n pair:

[Hqiv/Physics/DynamicNucleonPN.lean](#).

Caustic binding stack:

[Hqiv/Physics/NuclearCausticBinding.lean](#), [Hqiv/Physics/NuclearCurvatureBinding.lean](#).

Nuclear spectra module:

[Hqiv/Physics/NuclearAndAtomicSpectra.lean](#).

Isotope valley scaffold:

[Hqiv/Physics/HQIVNuclei.lean](#).

APPENDIX B. WITNESS SCRIPTS AND JSON EXPORTS

scripts/hqiv_isotope_stability_half-life.py:

Stability readout, half-life, outside support; writes data/isotope_stability_half-life.json.

scripts/hqiv_dynamic_beta_isotope.py:

Three-ledger β channel chart; data/dynamic_beta_isotope_chart.json.

scripts/hqiv_isotope_pdg_benchmark.py:

Light-panel benchmark table; data/isotope_pdg_benchmark.json.

scripts/hqiv_nuclear_outside_temperature_dynamics.py:

Outside $T(\xi)$ and gravity binding stack.

scripts/hqiv_bond_state_network.py:

Molecular bond-state network; T₂, T₂O hosts; molecular_host_phi_epsilon. Fano/Hopf bridge energy witness.

hqiv_lab/:

Chem & materials lab package: MaterialsLab, derived allotropes, hqiv-lab CLI; tests in hqiv_lab/tests/. Install: pip install -e . from repo root (pyproject.toml).

```

scripts/hqiv_phase_geometry_density.py:
    Legacy bridge: calls hqiv_lab for unit cells;  $\rho_{\text{curv}}$  melt readout.
scripts/hqiv_thermodynamic_phase_from_tp.py:
     $(T, P) \rightarrow$  derived phase; bulk scales use derived  $\rho_{\text{curv}}$ .
scripts/hqiv_homogeneous_curvature_feedback.py:
     $B_{\text{hom}} + \delta B$  self-consistent demo (optional chart path).
scripts/hqiv_phase_material_response.py:
     $n, \varepsilon_r, k_{\text{th}}, C_p, L_f, \eta, \Delta n$ .
scripts/test_hqiv_phase_material_response.py:
    H2O ice/liquid response witnesses.

Run from repository root (see papers/nucleon_binding/scripts/README.md):
pip install -e .
hqiv-lab H2O
hqiv-lab CH4 --json
python3 scripts/hqiv_isotope_pdg_benchmark.py --qualify-em-tipping
python3 scripts/hqiv_isotope_stability_half-life.py --A 1 --Z 0 \
    --qualify-em-tipping --lab-temperature-K 300 --gravity-stack full
    
```

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